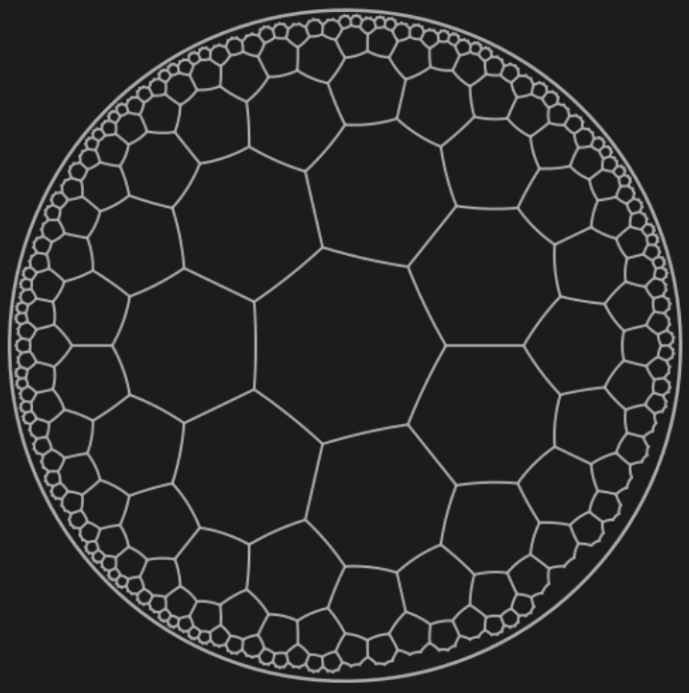
HYPERBOLIC GARDENS



A Coloring Book of Infinite Tilings

50 hand-finishable hyperbolic designs

MathRelaxing

HYPERBOLIC GARDENS

A Coloring Book of Infinite Tilings

50 Poincare-disk {p,q} tessellations · MathRelaxing

About this book

Every design in these pages is a tiling of the hyperbolic plane — a world where space curves away from itself so steeply that infinitely many identical shapes fit inside a single circle. They are drawn in the Poincare disk, the same model of hyperbolic geometry that inspired M.C. Escher's famous Circle Limit woodcuts.

Each pattern is named by its Schlafli symbol {p, q}: it is built from regular p-sided polygons, with exactly q of them meeting around every corner. The tiles appear to shrink toward the rim, but in hyperbolic space they are all exactly the same size — the pattern truly continues forever, never reaching the edge.

The next page explains the geometry behind the drawings. After that, 50 unique tilings await your colour.

How to colour:

- Any pens, pencils or markers work on the matte paper.
- Try colouring by ring, by tile size, or by symmetry — each choice reveals a different hidden structure.
- Slip a spare sheet behind the page if you use wet markers.

Enjoy the infinite.

The mathematics

The Poincare disk. Every design lives inside one circle — a map of the entire hyperbolic plane. The straight lines of this world, its geodesics, appear as arcs of circles that meet the boundary at right angles. That is why every edge in these tilings curves so gracefully.

The Schlafli symbol. Each tiling is described by two numbers: regular p-sided polygons, with q of them meeting at every vertex.

$$\{\, p,\ q \,\}$$

When is it hyperbolic? A regular tiling is spherical, flat, or hyperbolic depending only on p and q. It curves into the hyperbolic plane exactly when

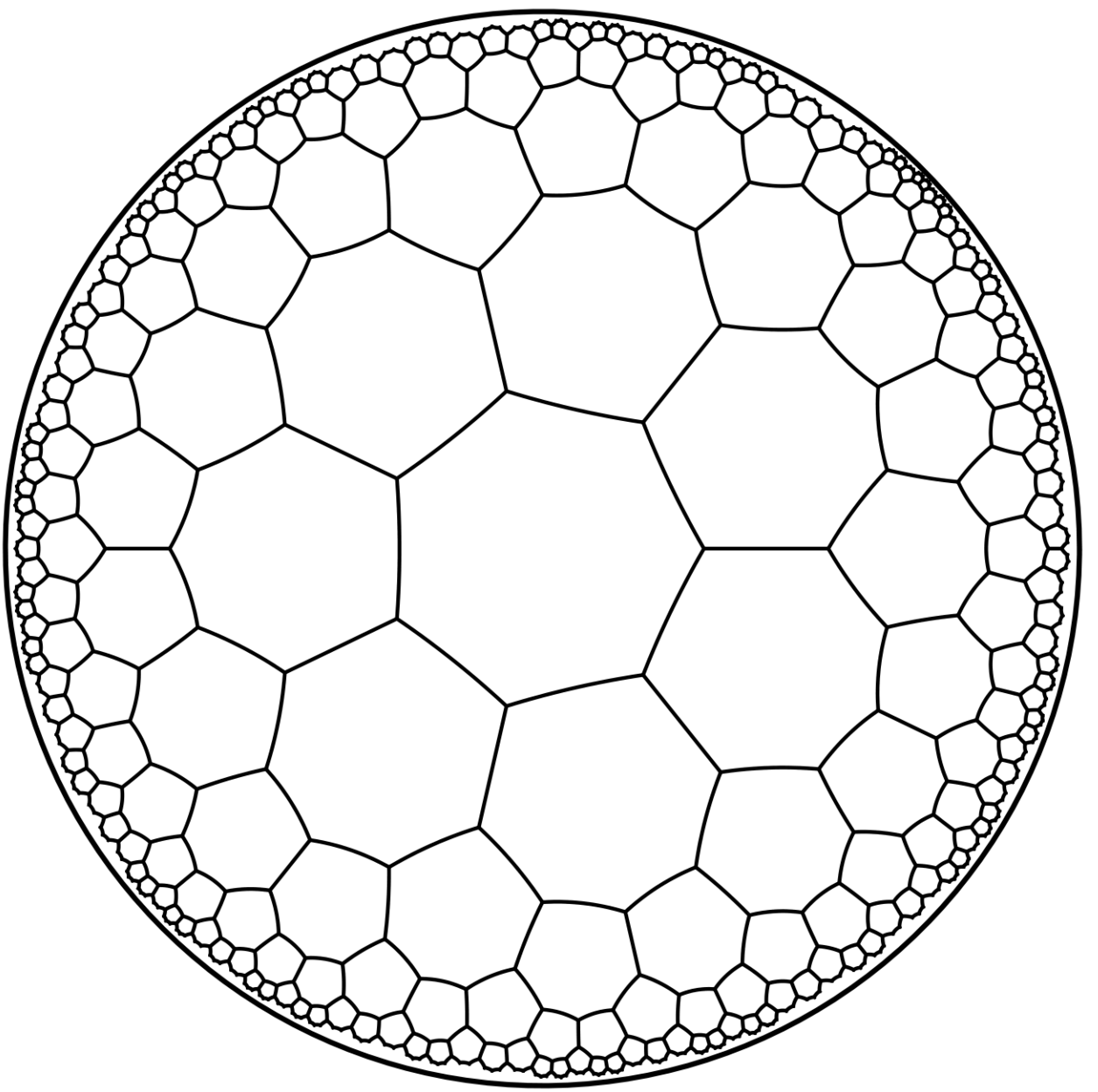
$$\frac{1}{p} + \frac{1}{q} < \frac{1}{2}$$

If the sum is greater than one half the tiling closes up on a sphere (only five exist); if it equals one half the tiling is flat (only three). Below one half there are infinitely many — and every one of them lives here.

The size of a tile. In the disk each polygon has the same hyperbolic circumradius R, fixed by p and q through

$$\cosh R = \frac{\cos(\pi/q)}{\sin(\pi/p)}$$

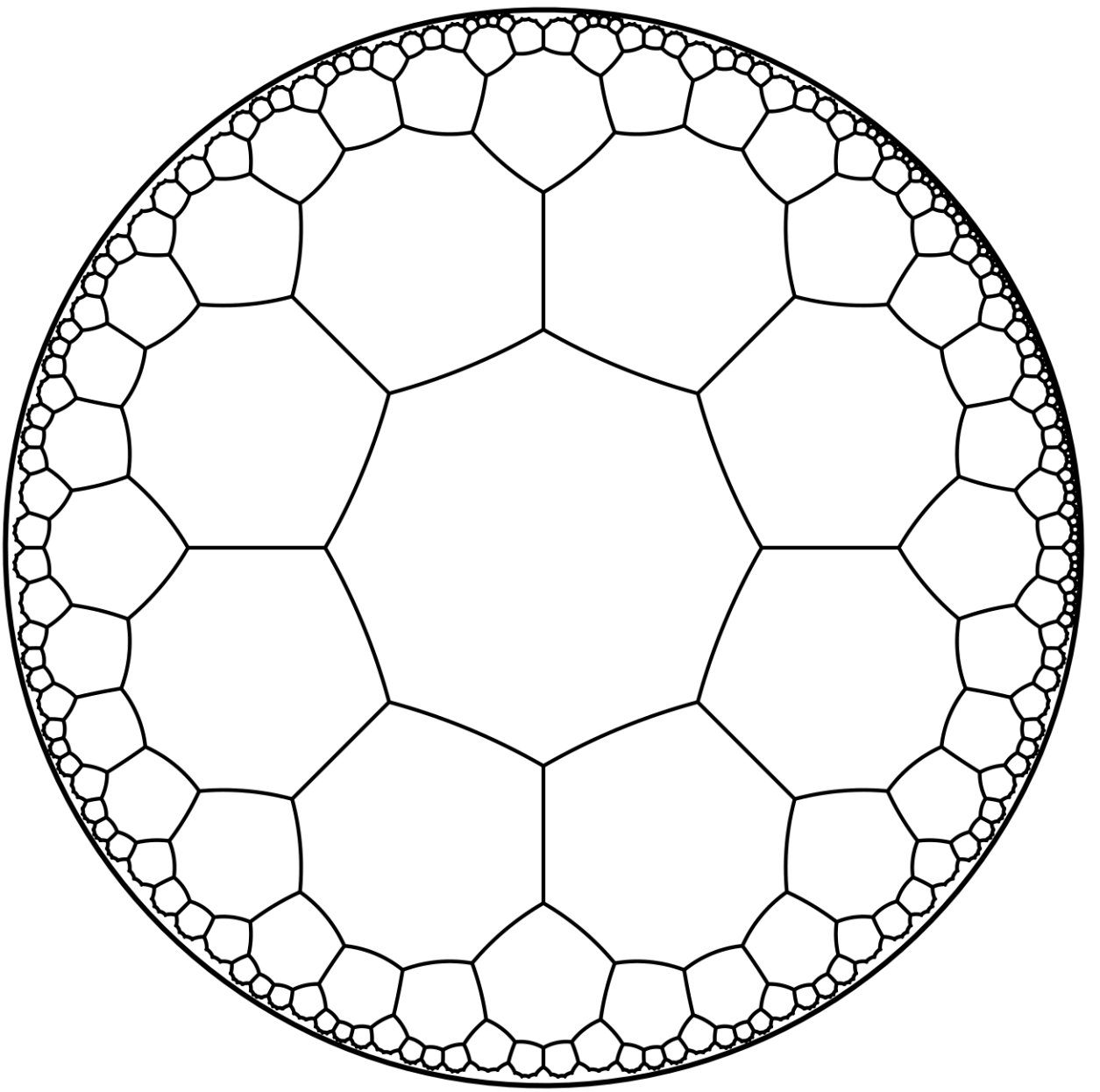
Building the pattern. Starting from a single tile, we reflect it across its edges again and again. These reflections form a hyperbolic symmetry group, and their orbit fills the disk with the endless garden you colour.



{7, 3}

Regular heptagons · 3 meeting at every vertex

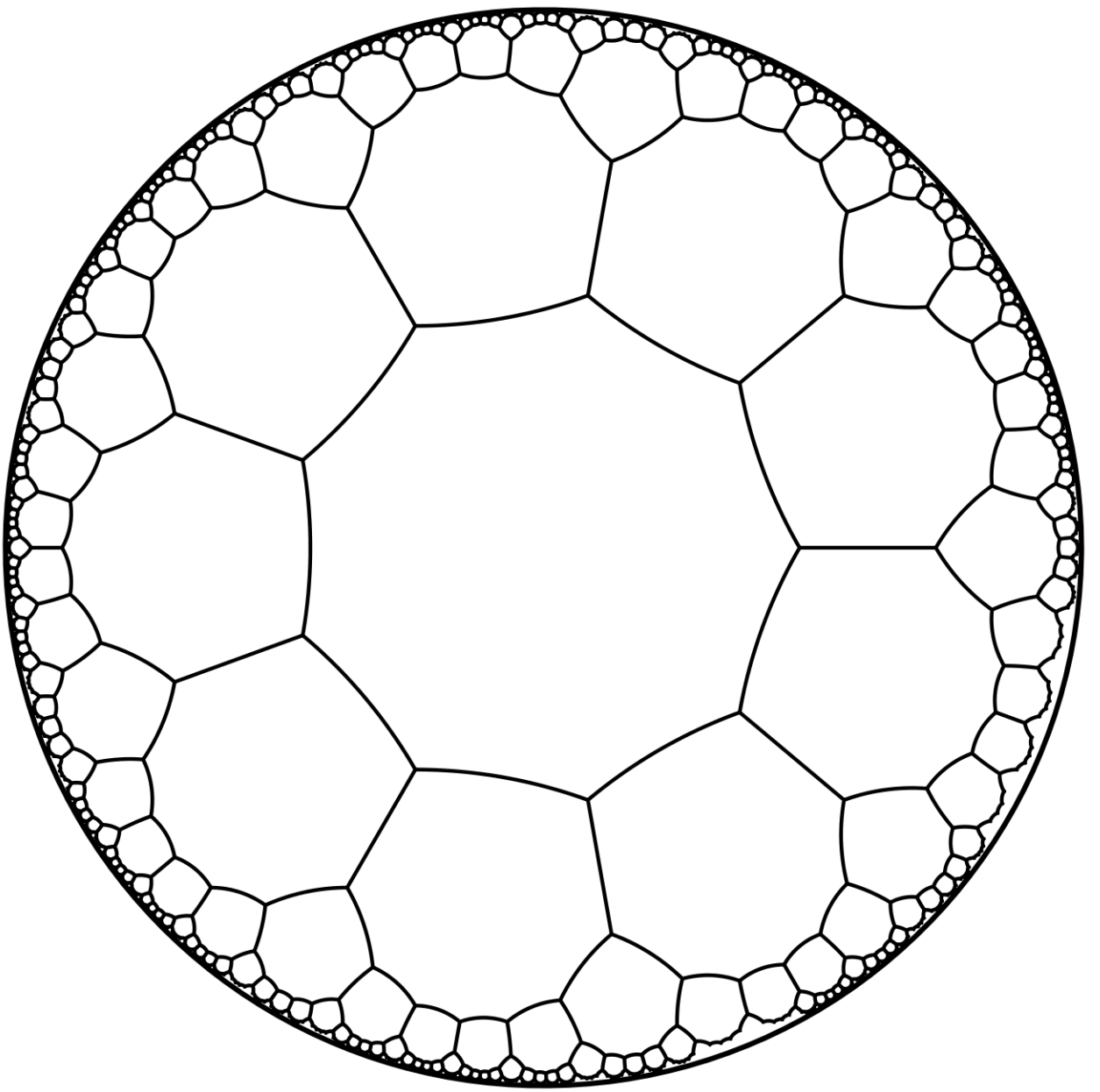
$\frac{1}{7} + \frac{1}{3} \approx 0.476 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/3)}{\sin(\pi/7)} \approx 1.15$



{8, 3}

Regular octagons · 3 meeting at every vertex

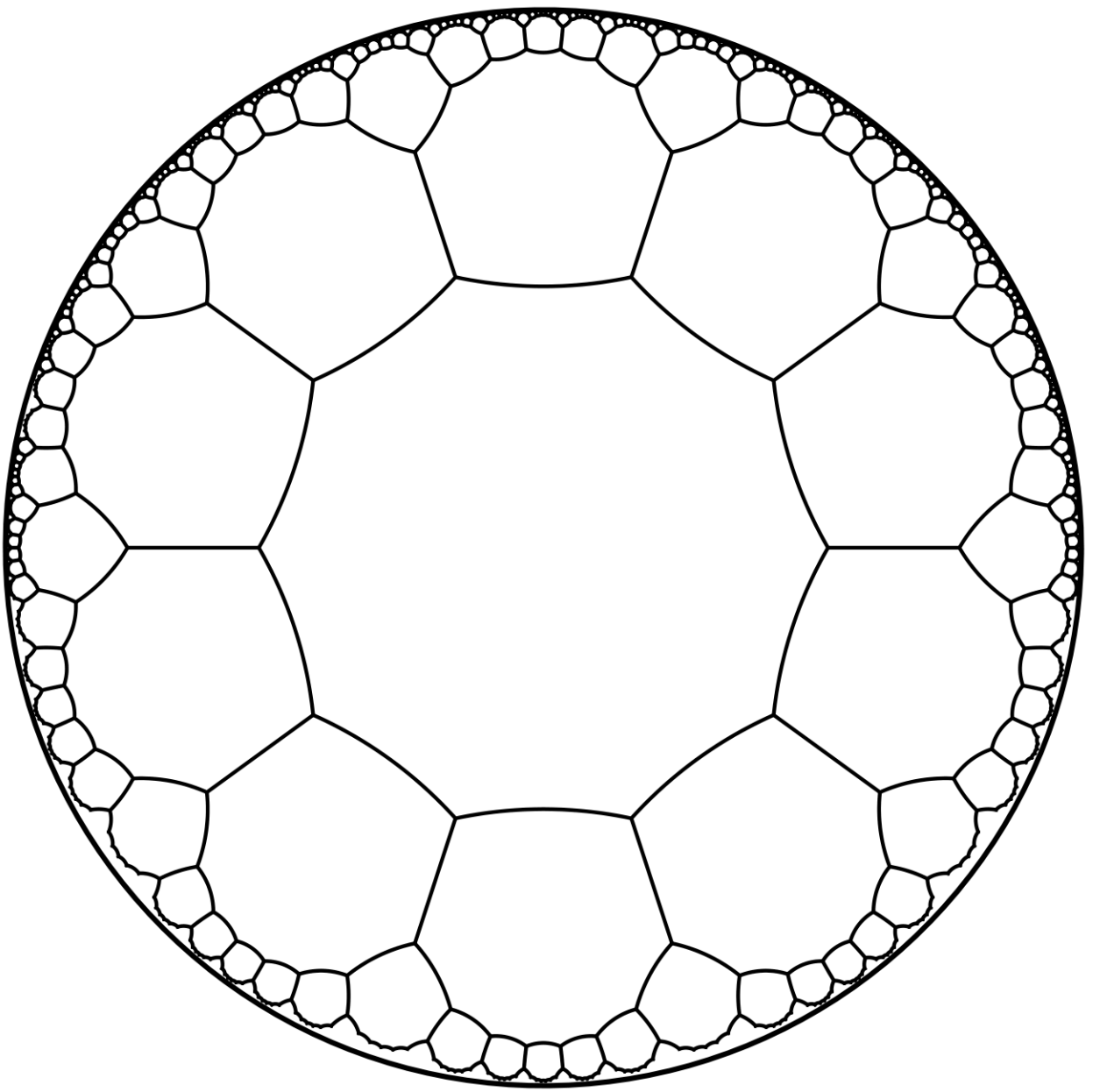
$\frac{1}{8} + \frac{1}{3} \approx 0.458 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/3)}{\sin(\pi/8)} \approx 1.31$



{9, 3}

Regular nonagons · 3 meeting at every vertex

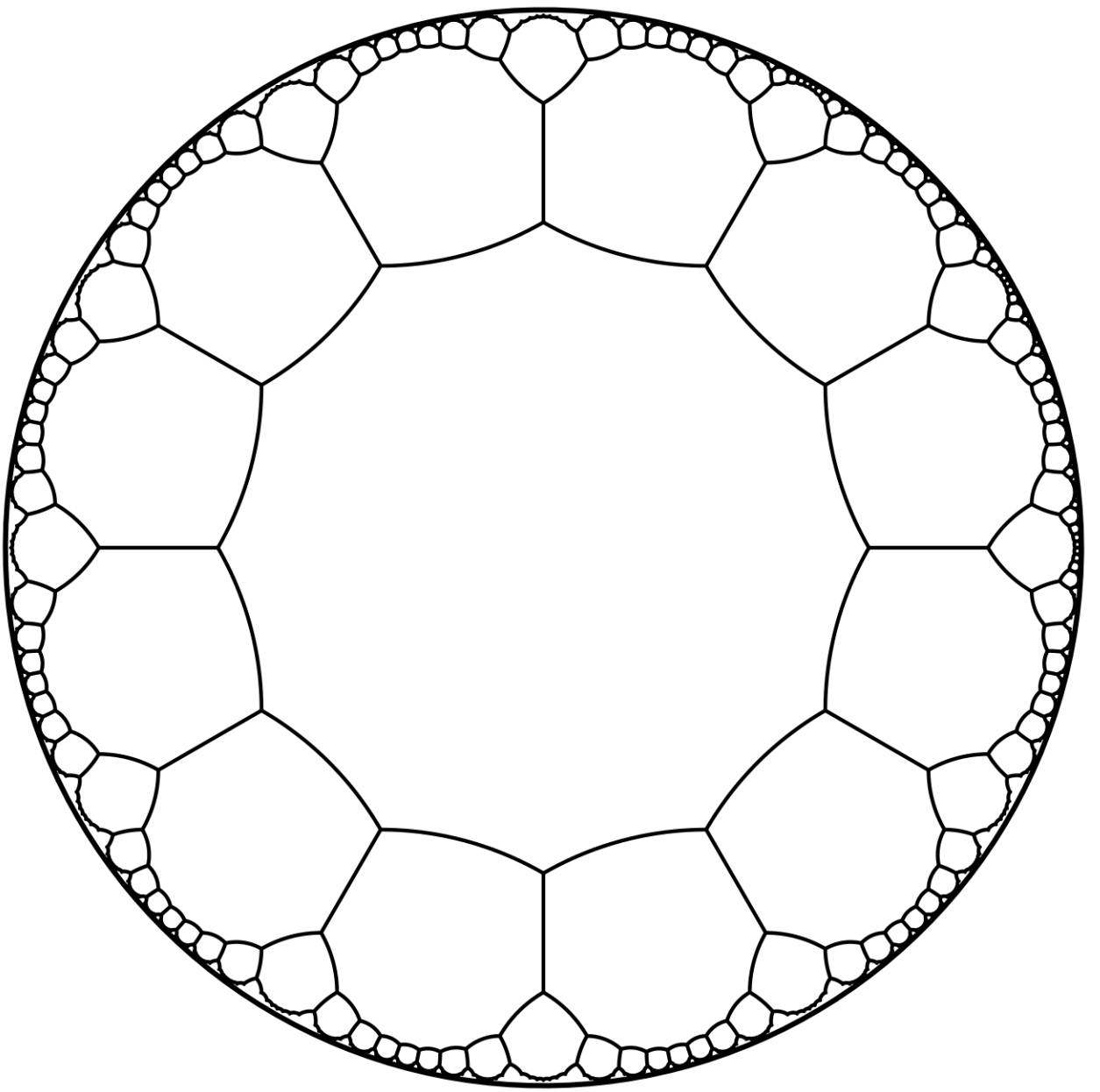
$\frac{1}{9} + \frac{1}{3} \approx 0.444 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/3)}{\sin(\pi/9)} \approx 1.46$



{10, 3}

Regular decagons · 3 meeting at every vertex

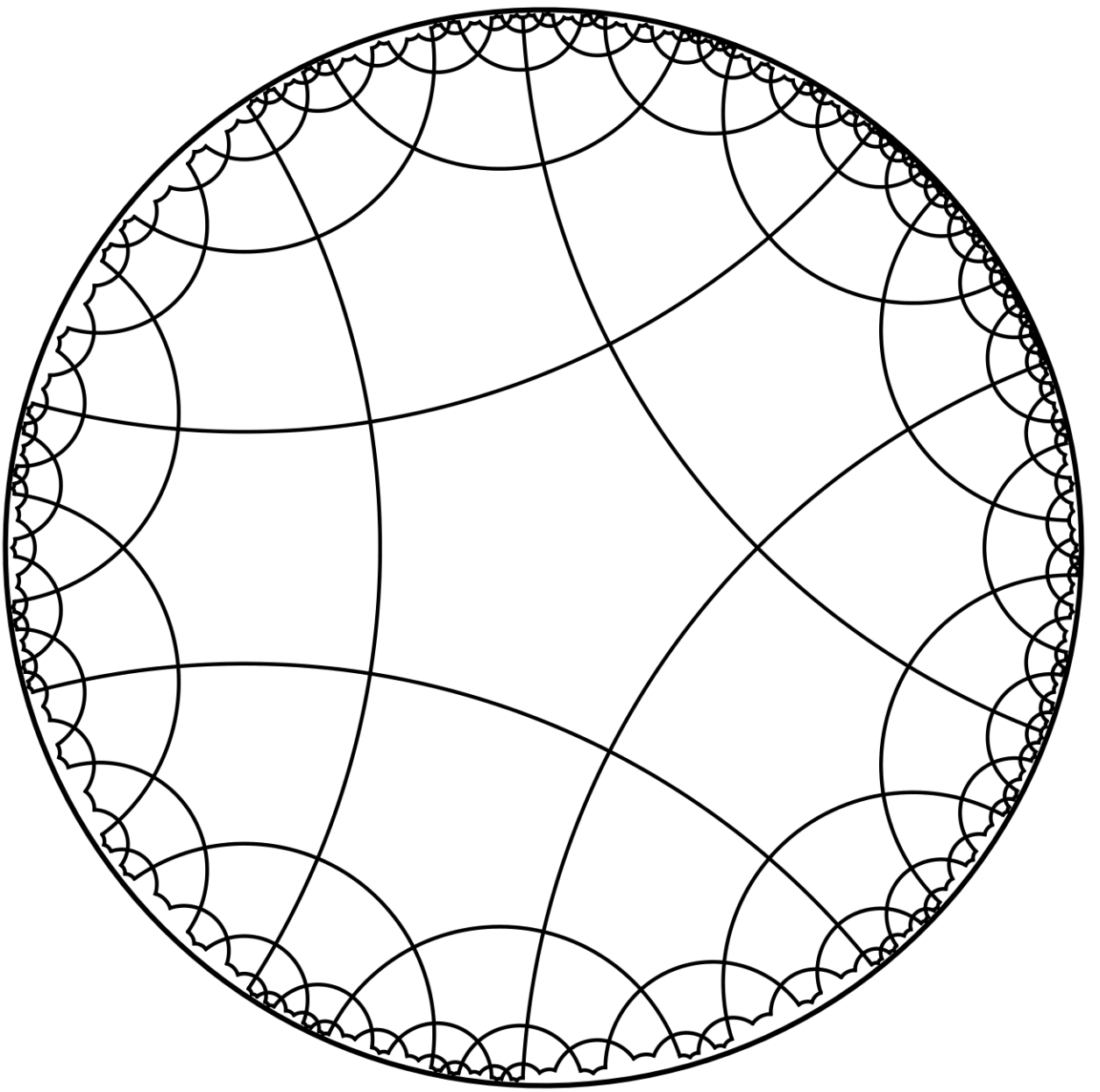
$\frac{1}{10} + \frac{1}{3} \approx 0.433 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/3)}{\sin(\pi/10)} \approx 1.62$



{12, 3}

Regular dodecagons · 3 meeting at every vertex

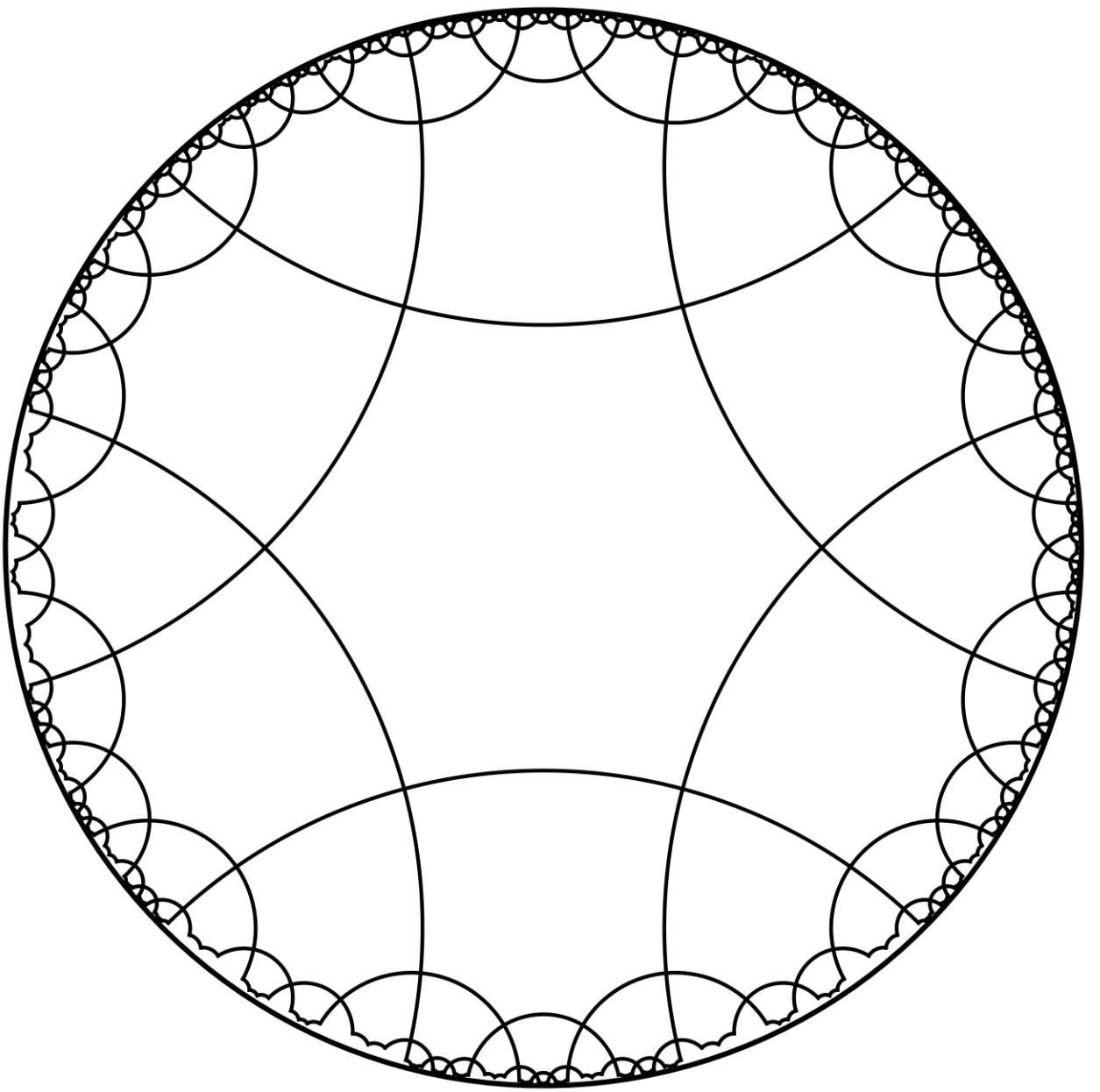
$\frac{1}{12} + \frac{1}{3} \approx 0.417 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/3)}{\sin(\pi/12)} \approx 1.93$



{5, 4}

Regular pentagons · 4 meeting at every vertex

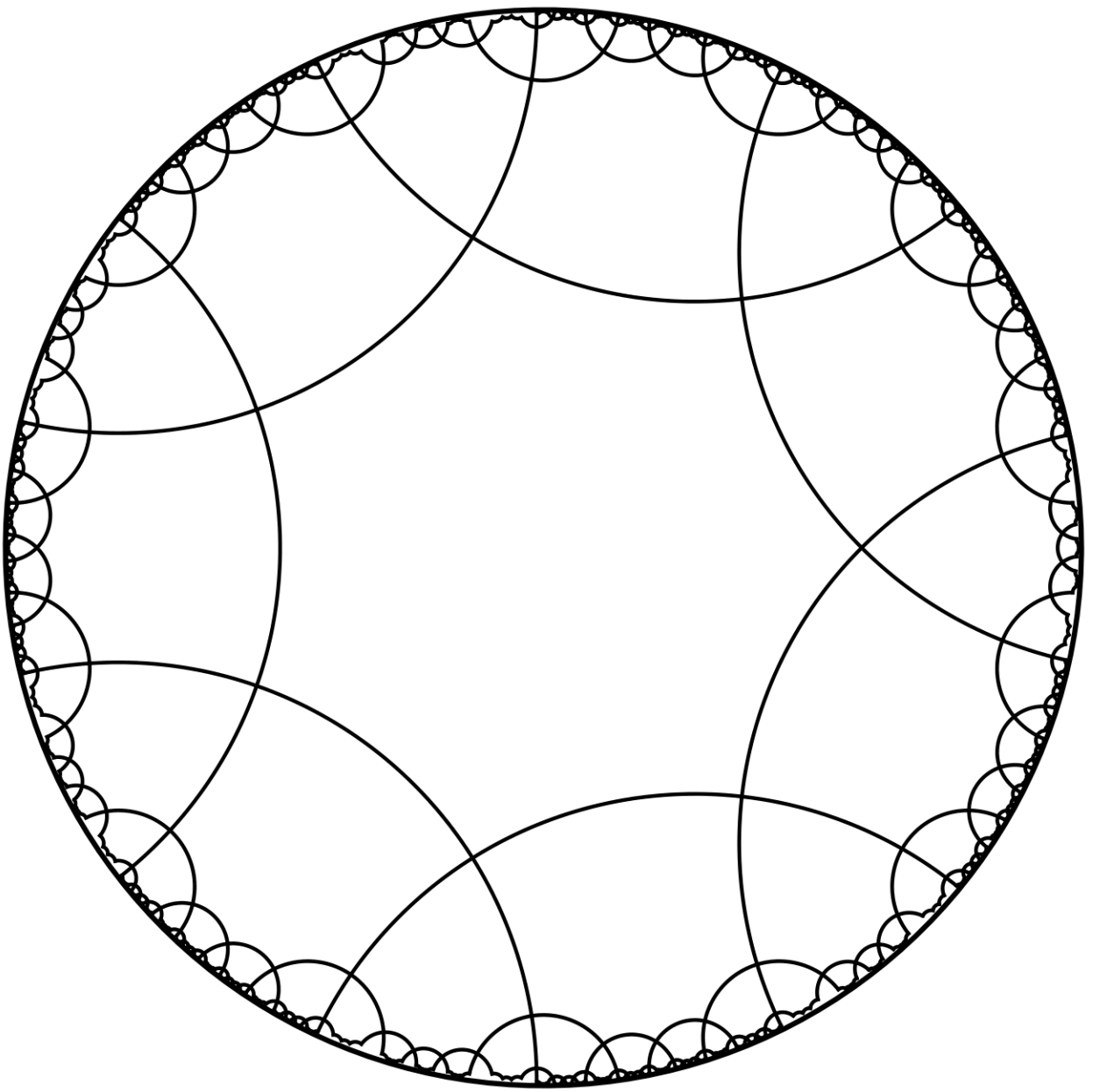
$\frac{1}{5} + \frac{1}{4} \approx 0.450 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/4)}{\sin(\pi/5)} \approx 1.20$



{6, 4}

Regular hexagons · 4 meeting at every vertex

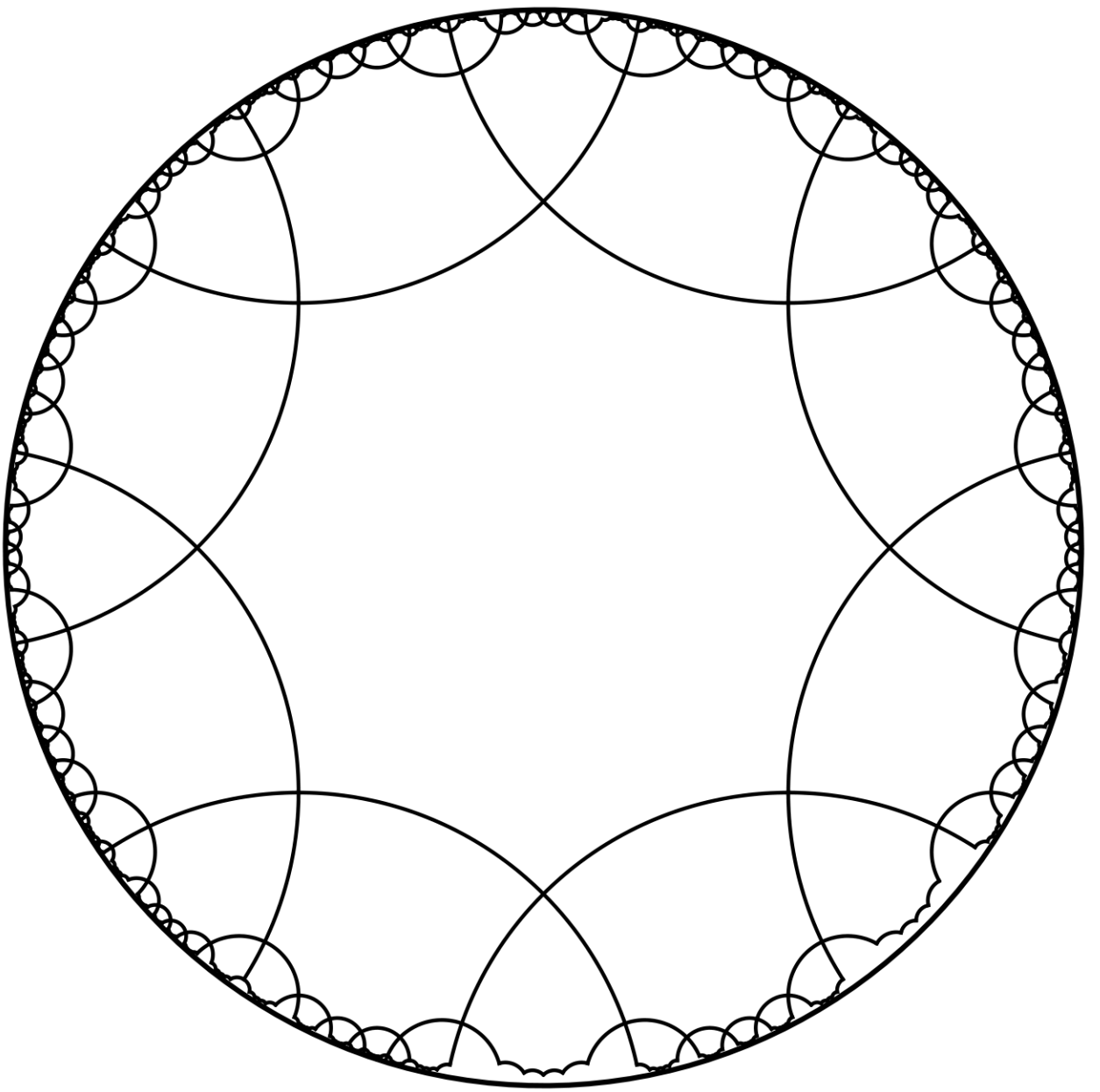
$\frac{1}{6} + \frac{1}{4} \approx 0.417 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/4)}{\sin(\pi/6)} \approx 1.41$



{7, 4}

Regular heptagons · 4 meeting at every vertex

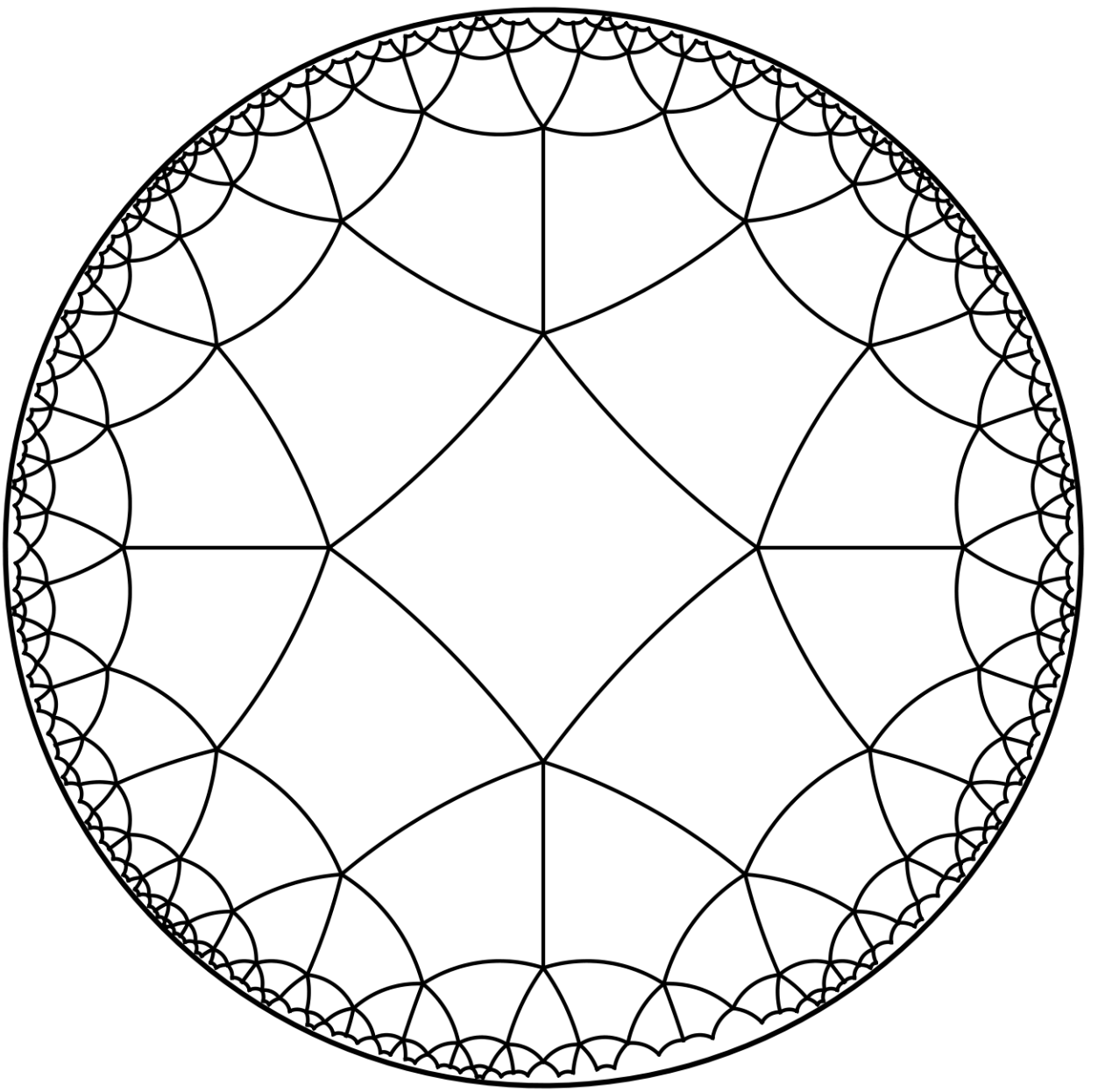
$\frac{1}{7}+\frac{1}{4}\approx 0.393<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/4)}{\sin(\pi/7)}\approx 1.63$



{8, 4}

Regular octagons · 4 meeting at every vertex

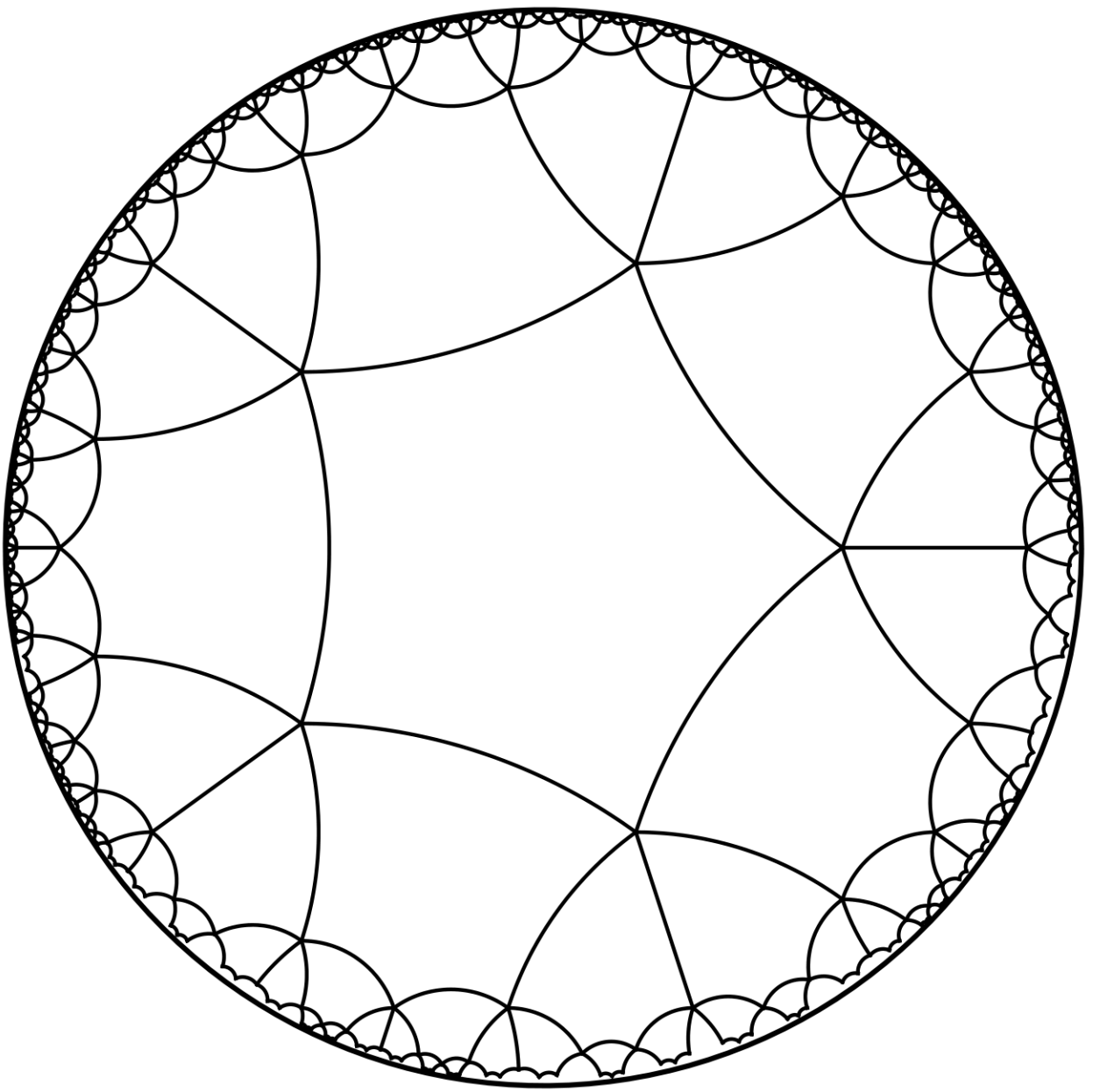
$\frac{1}{8} + \frac{1}{4} \approx 0.375 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/4)}{\sin(\pi/8)} \approx 1.85$



{4, 5}

Regular squares · 5 meeting at every vertex

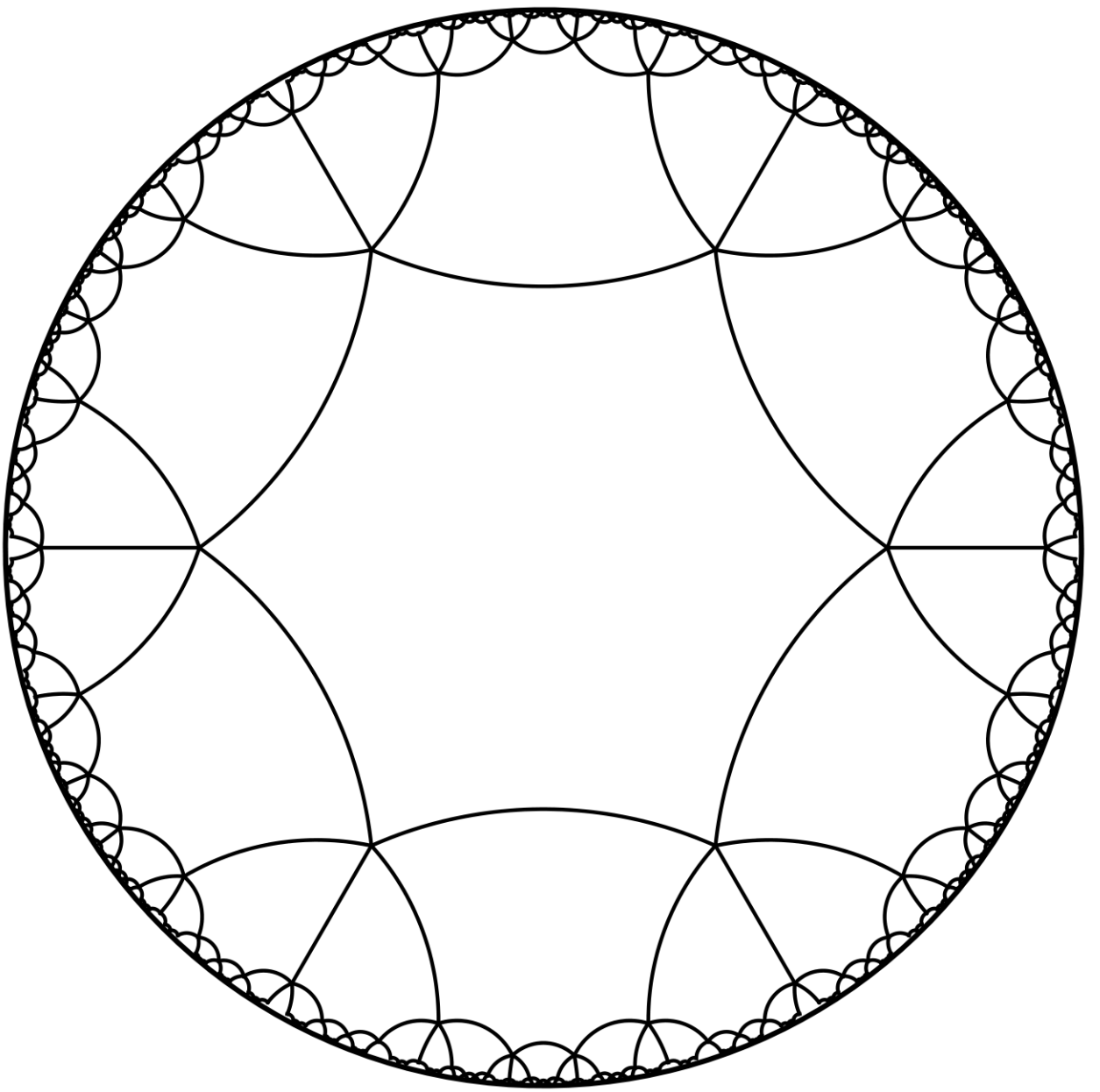
$\frac{1}{4} + \frac{1}{5} \approx 0.450 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/5)}{\sin(\pi/4)} \approx 1.14$



{5, 5}

Regular pentagons · 5 meeting at every vertex

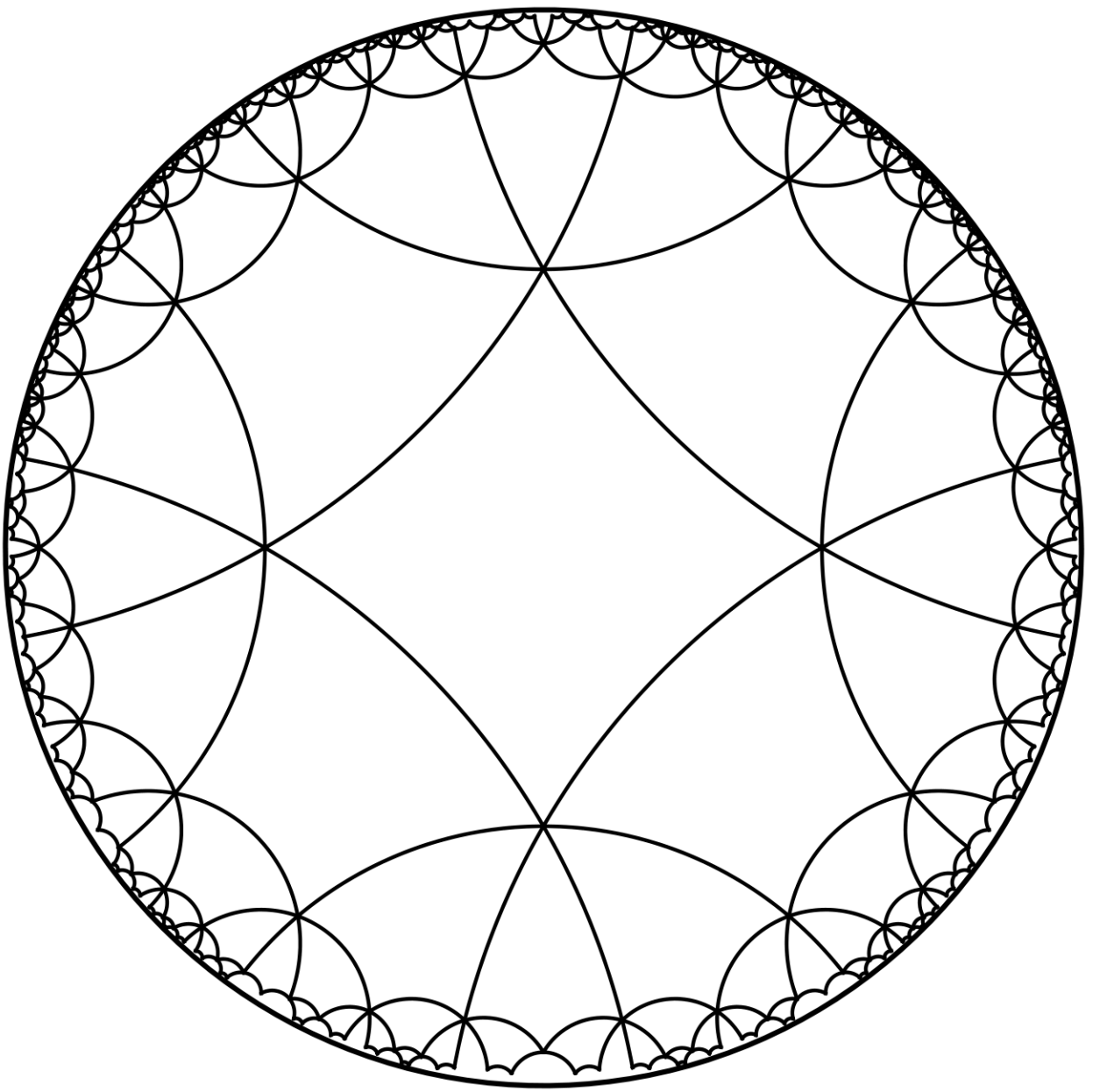
$\frac{1}{5} + \frac{1}{5} \approx 0.400 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/5)}{\sin(\pi/5)} \approx 1.38$



{6, 5}

Regular hexagons · 5 meeting at every vertex

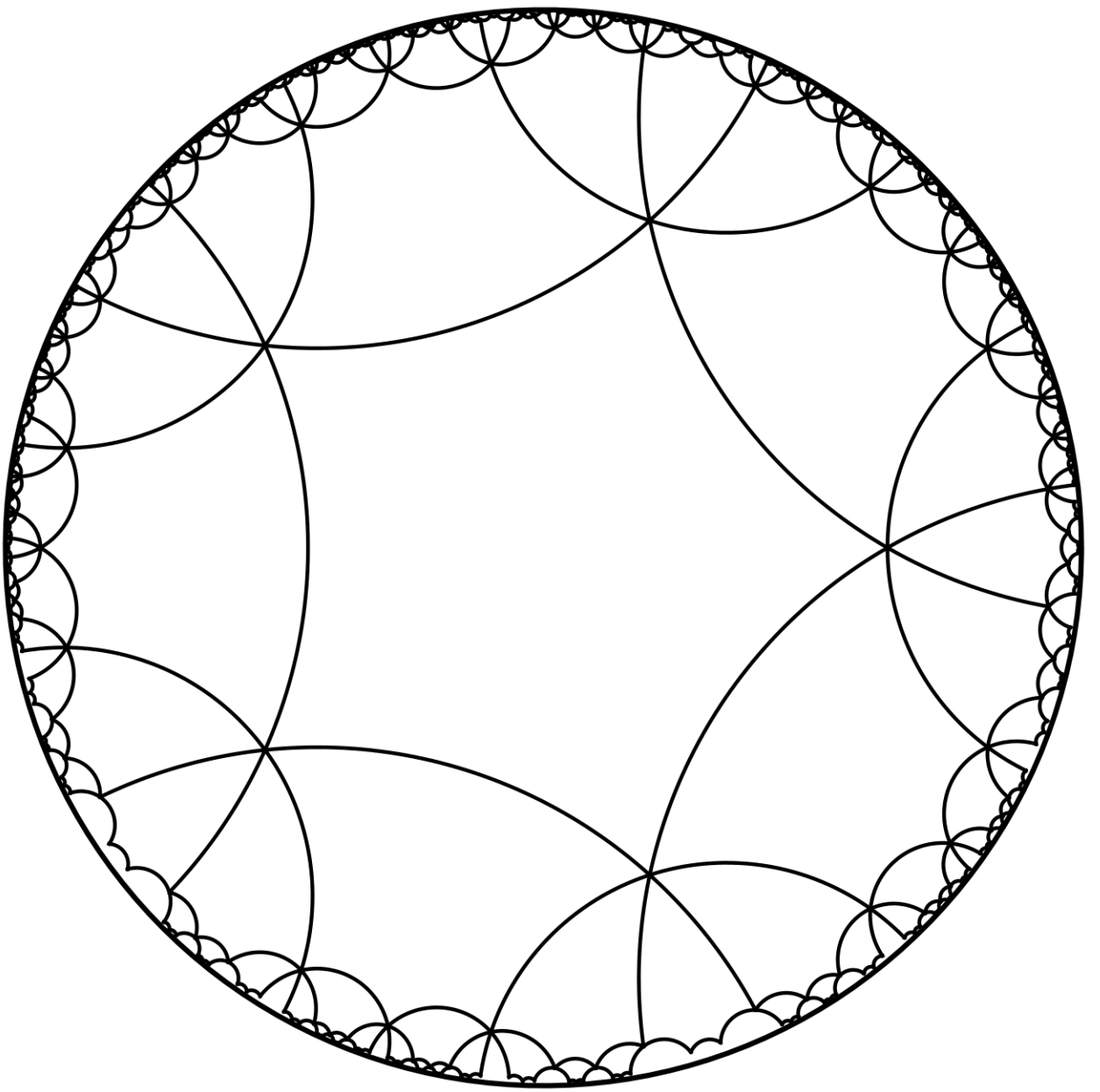
$\frac{1}{6} + \frac{1}{5} \approx 0.367 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/5)}{\sin(\pi/6)} \approx 1.62$



{4, 6}

Regular squares · 6 meeting at every vertex

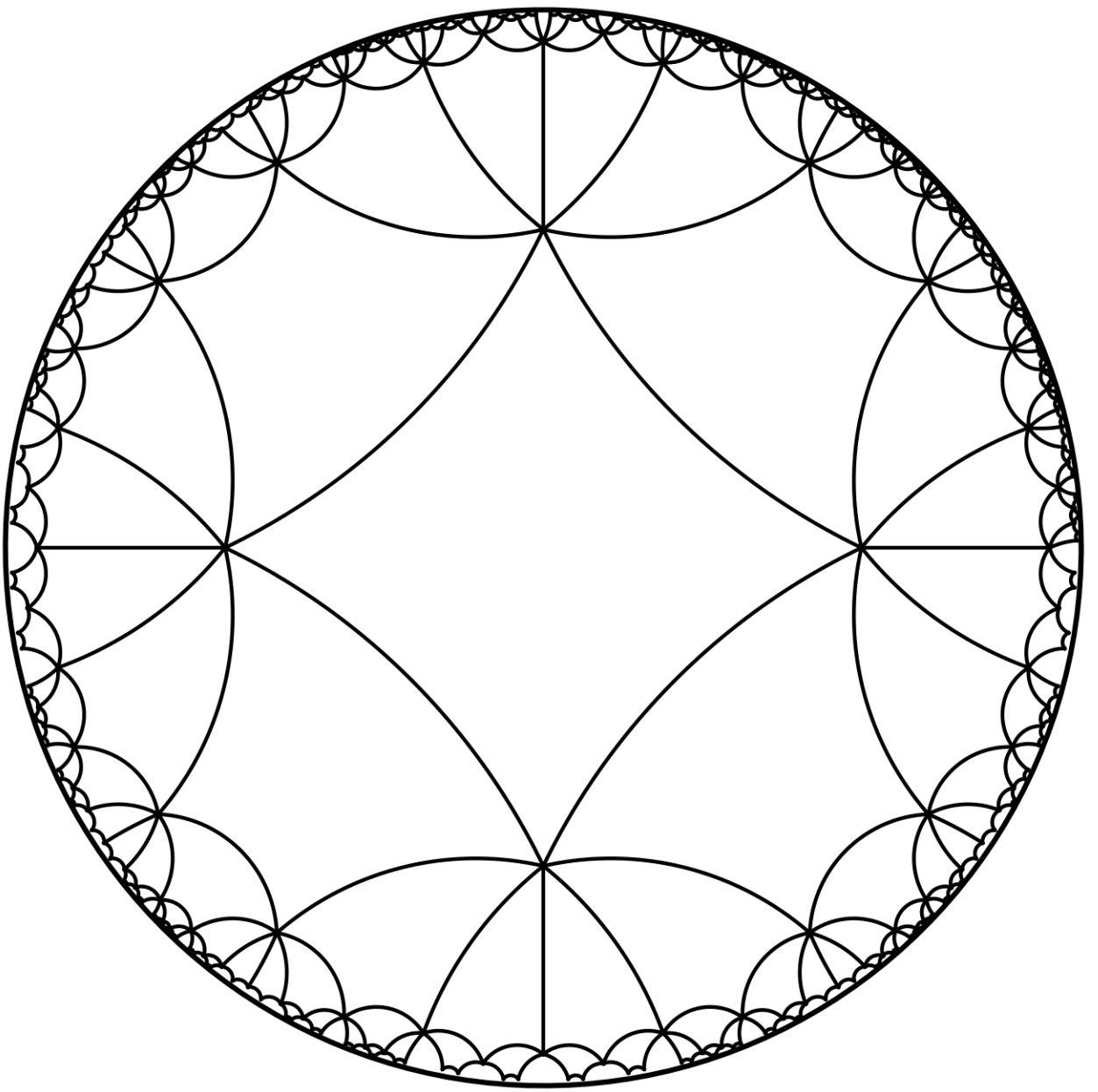
$\frac{1}{4} + \frac{1}{6} \approx 0.417 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/6)}{\sin(\pi/4)} \approx 1.22$



{5, 6}

Regular pentagons · 6 meeting at every vertex

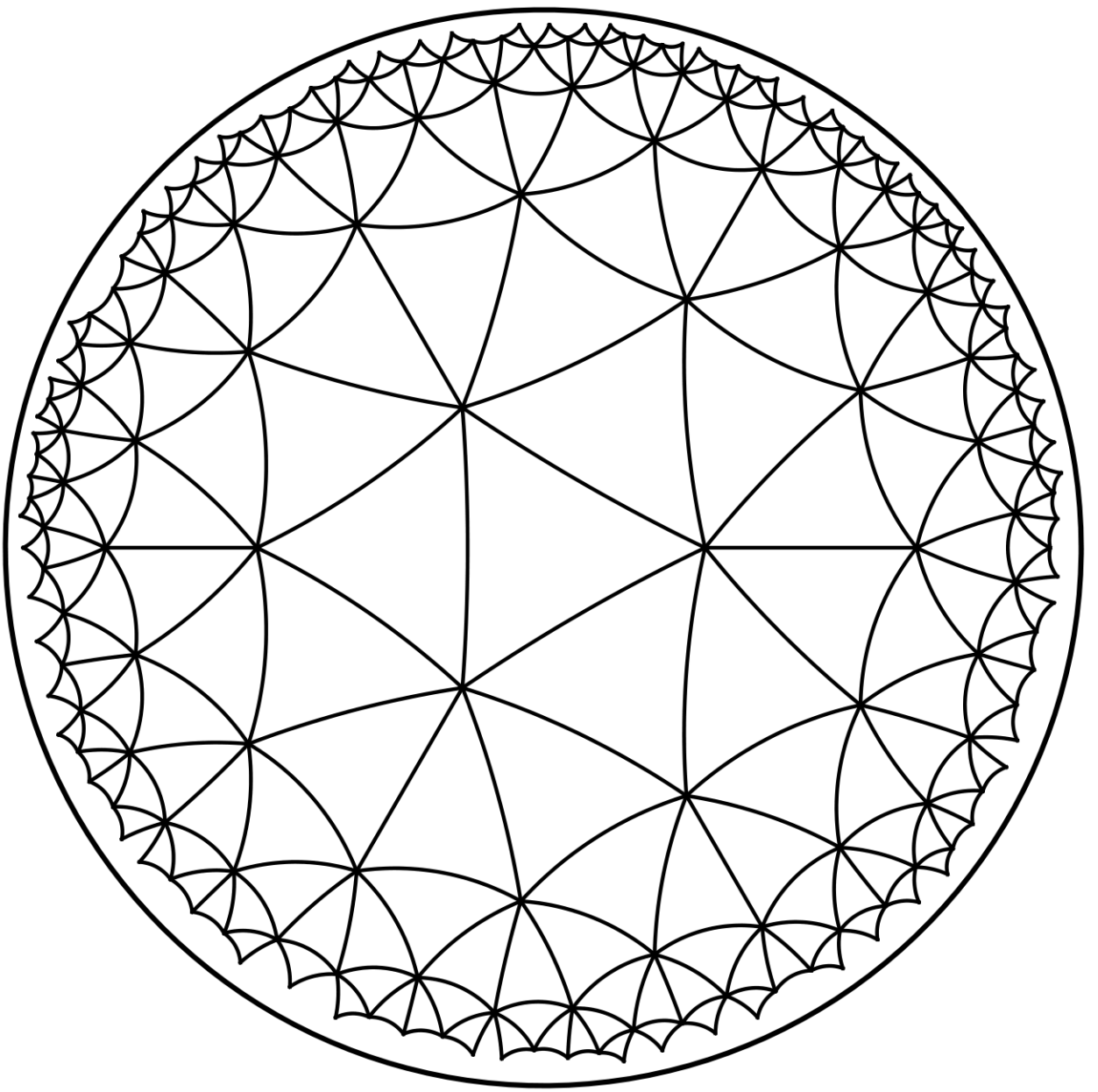
$\frac{1}{5} + \frac{1}{6} \approx 0.367 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/6)}{\sin(\pi/5)} \approx 1.47$



{4, 7}

Regular squares · 7 meeting at every vertex

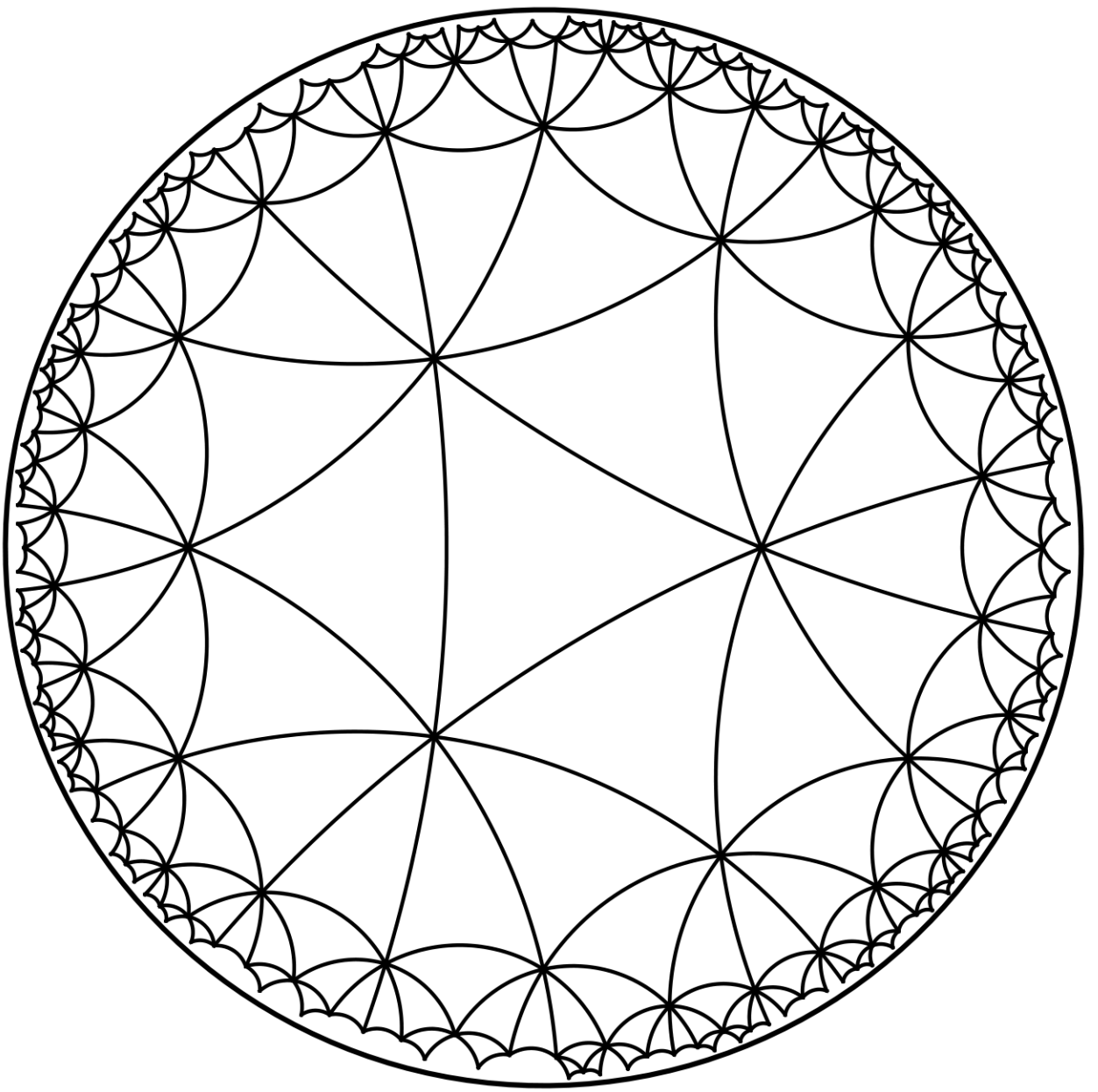
$\frac{1}{4} + \frac{1}{7} \approx 0.393 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/7)}{\sin(\pi/4)} \approx 1.27$



{3, 7}

Regular triangles · 7 meeting at every vertex

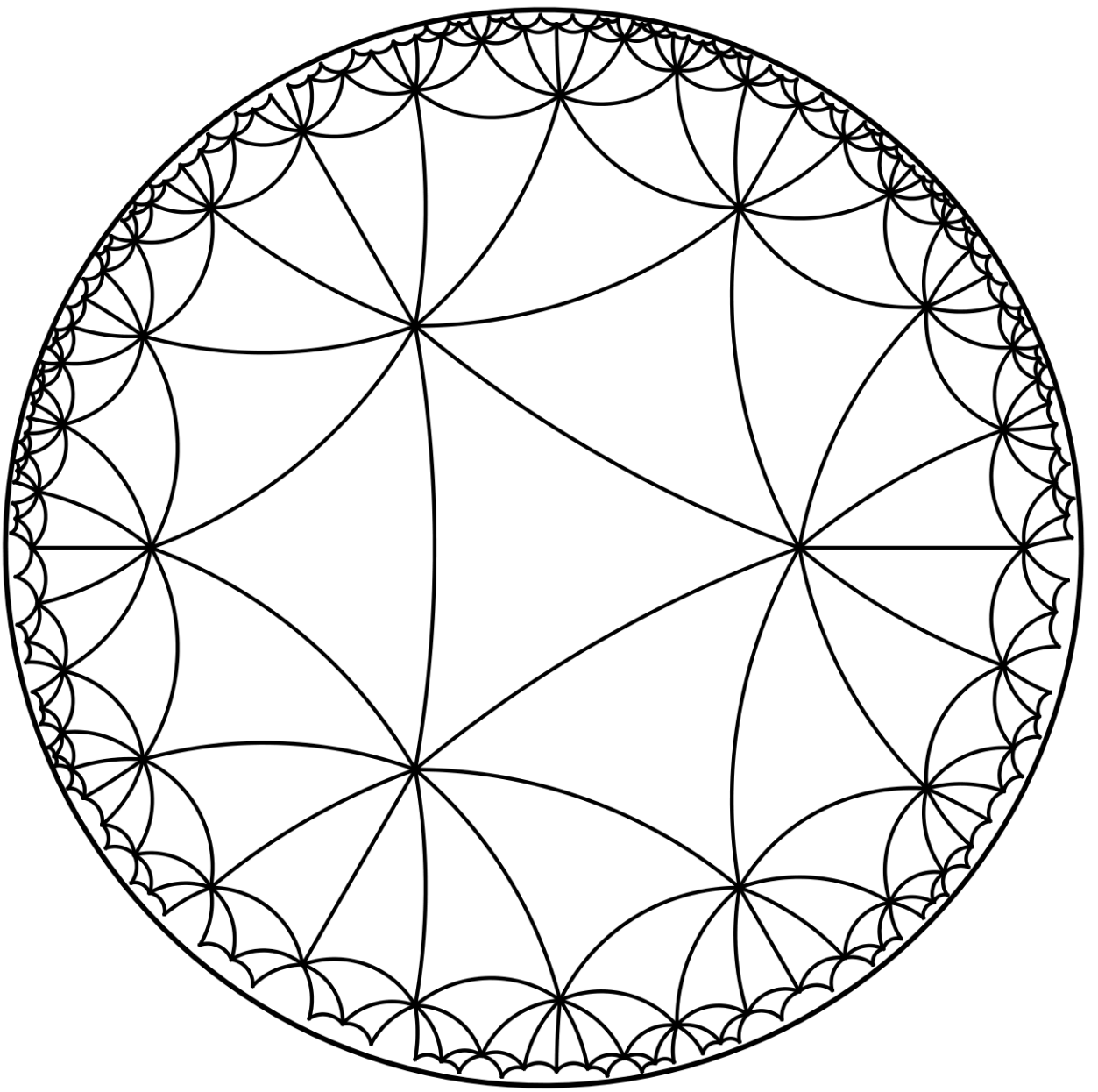
$\frac{1}{3} + \frac{1}{7} \approx 0.476 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/7)}{\sin(\pi/3)} \approx 1.04$



{3, 8}

Regular triangles · 8 meeting at every vertex

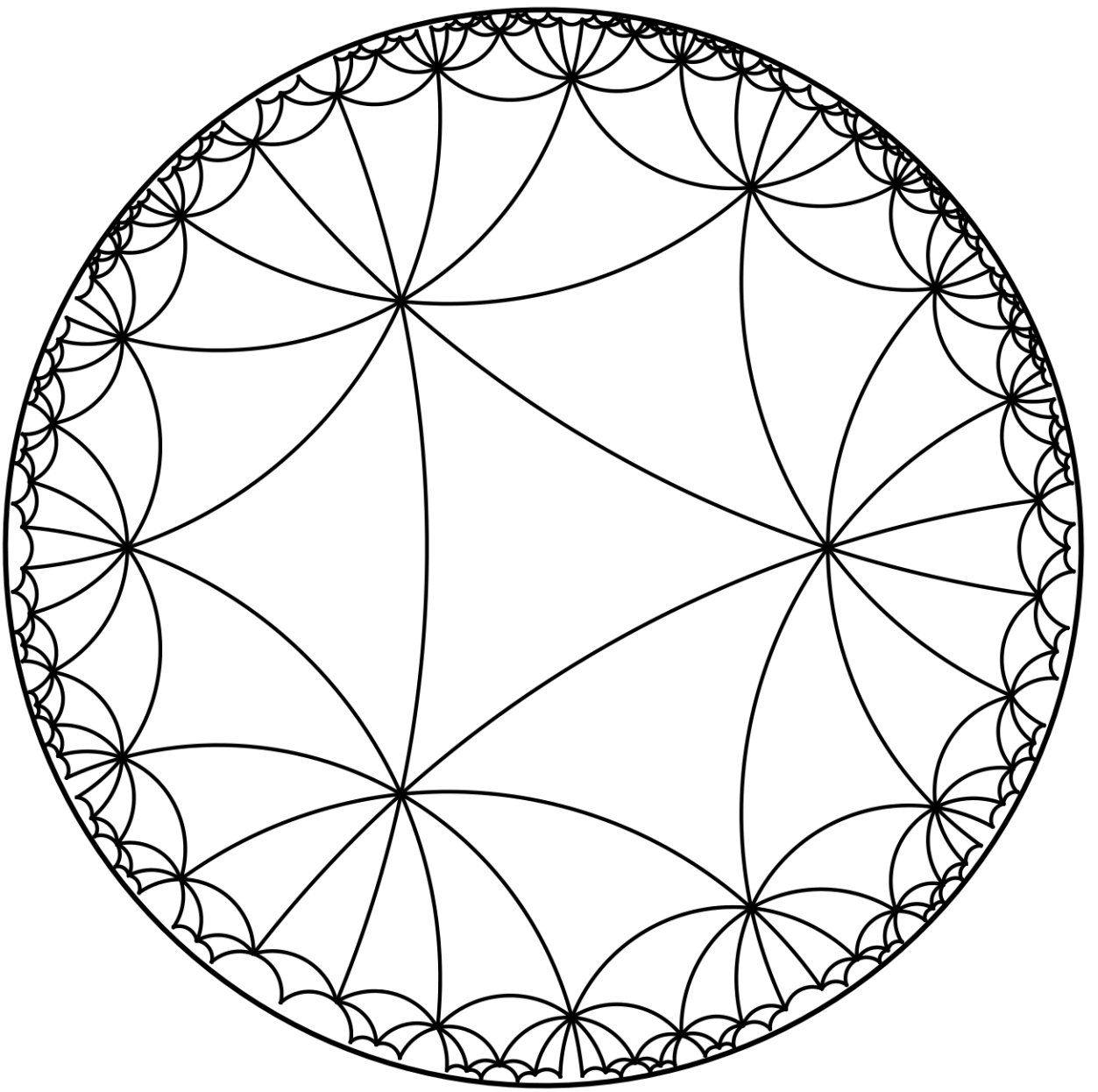
$\frac{1}{3} + \frac{1}{8} \approx 0.458 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/8)}{\sin(\pi/3)} \approx 1.07$



{3, 9}

Regular triangles · 9 meeting at every vertex

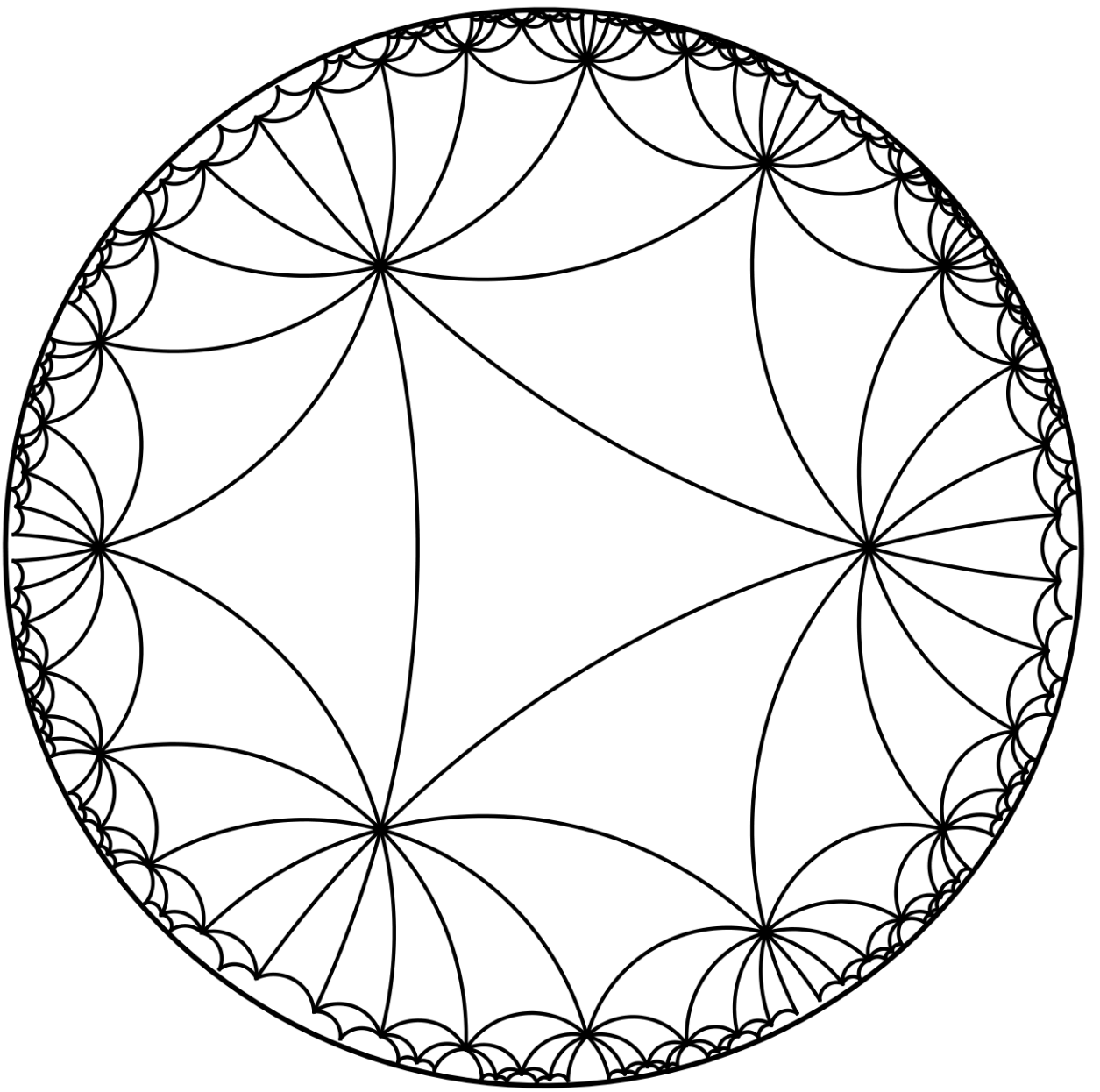
$\frac{1}{3} + \frac{1}{9} \approx 0.444 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/9)}{\sin(\pi/3)} \approx 1.09$



{3, 10}

Regular triangles · 10 meeting at every vertex

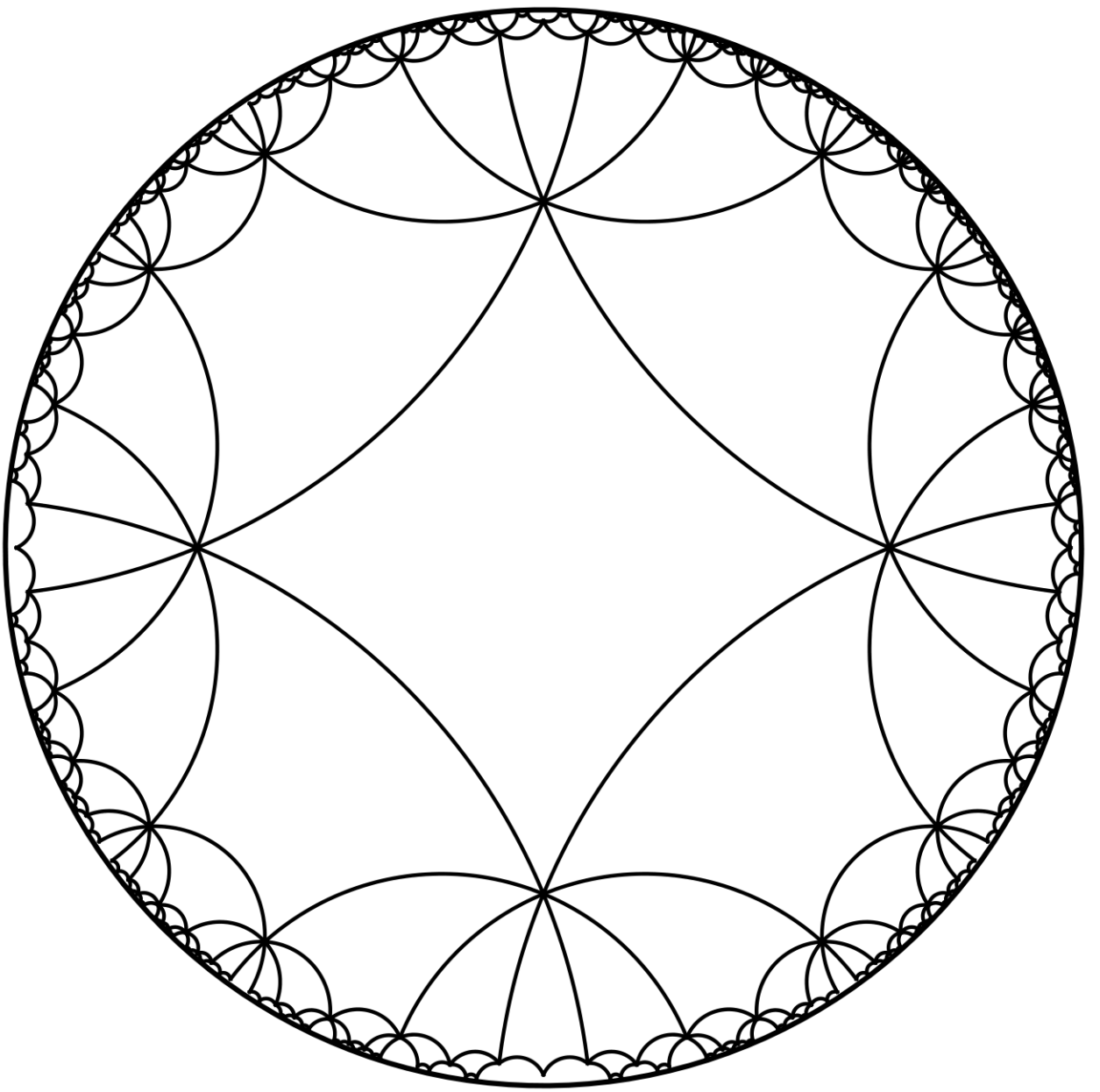
$\frac{1}{3} + \frac{1}{10} \approx 0.433 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/10)}{\sin(\pi/3)} \approx 1.10$



{3, 12}

Regular triangles · 12 meeting at every vertex

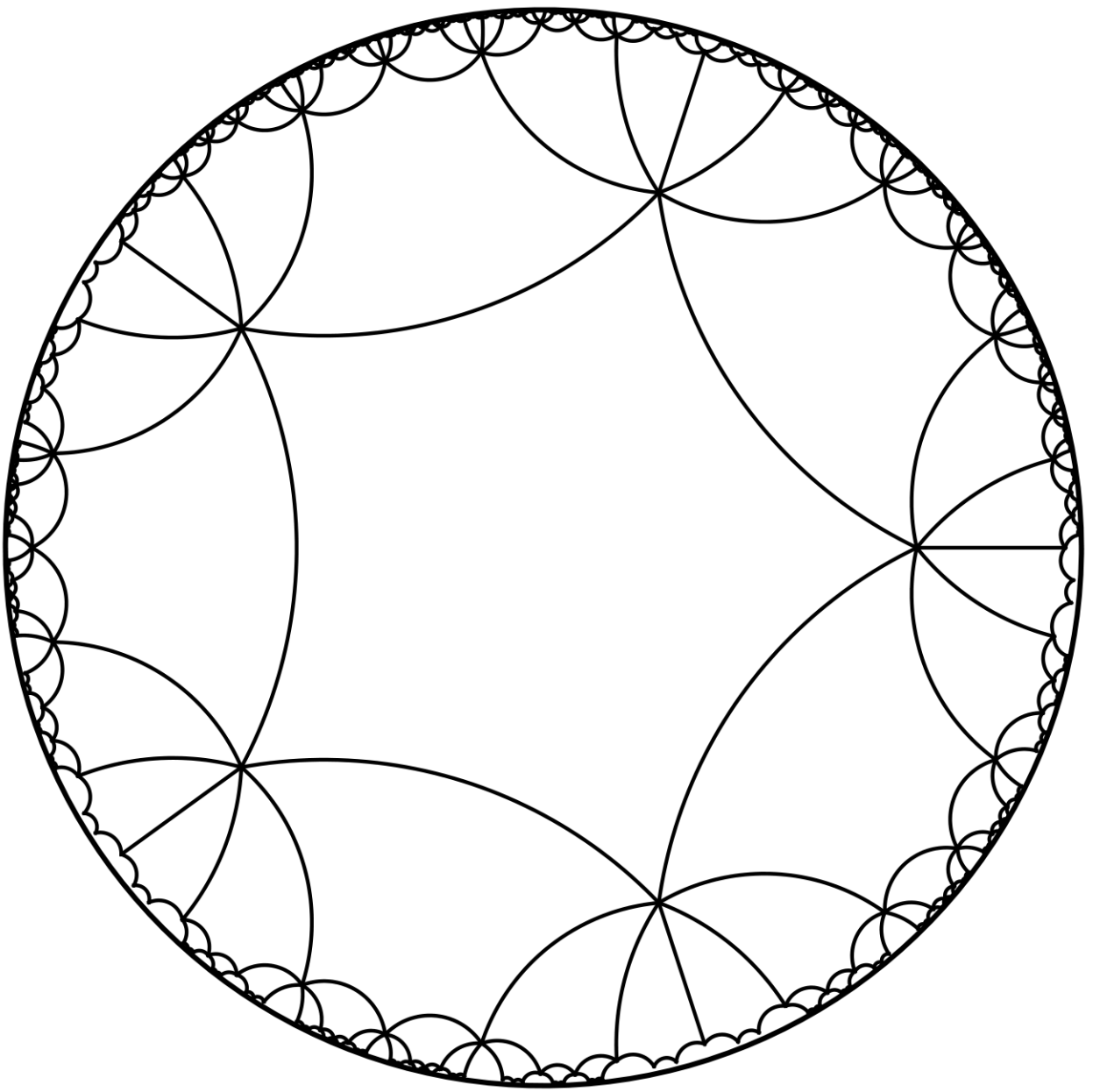
$\frac{1}{3} + \frac{1}{12} \approx 0.417 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/12)}{\sin(\pi/3)} \approx 1.12$



{4, 8}

Regular squares · 8 meeting at every vertex

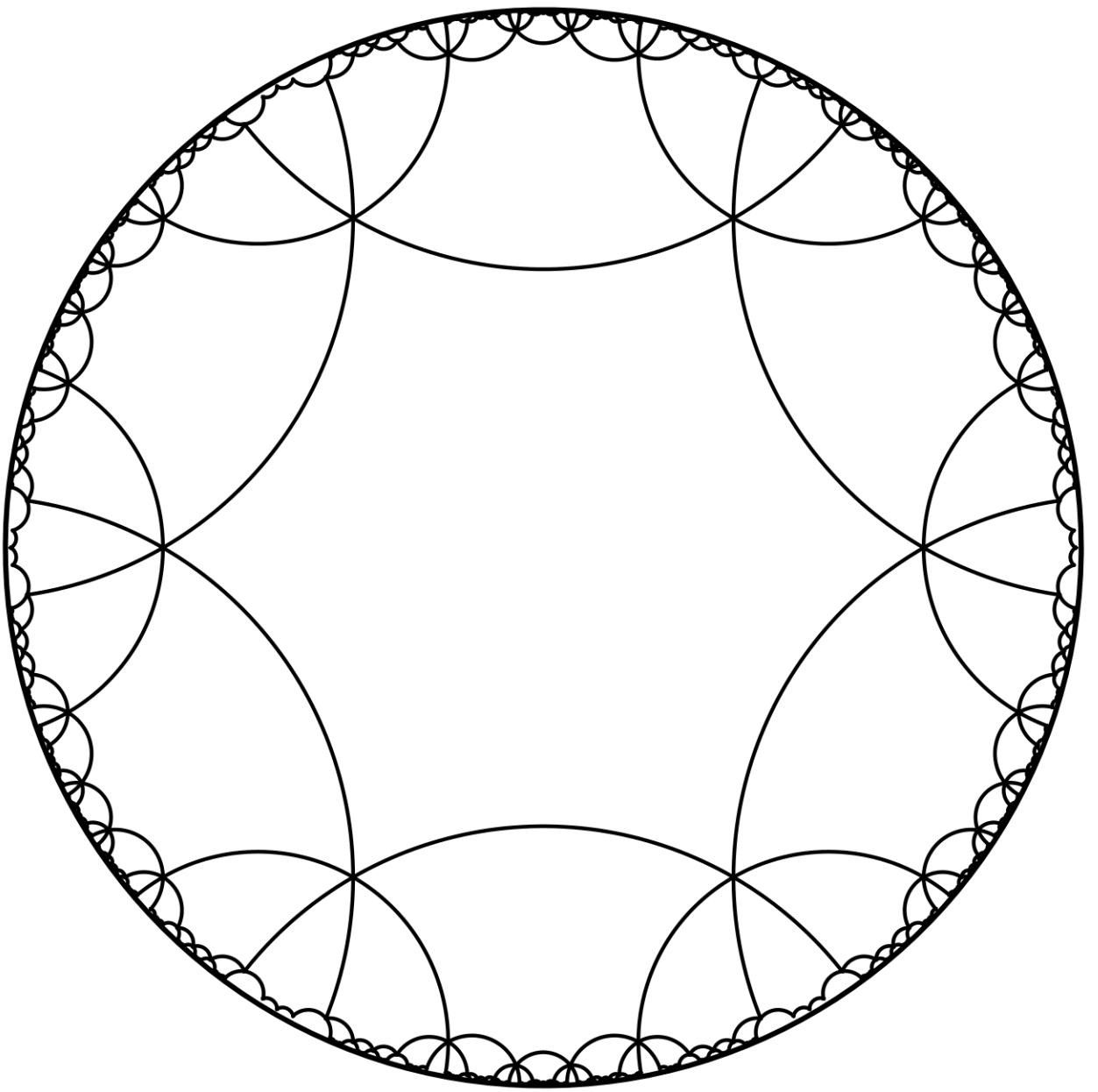
$\frac{1}{4} + \frac{1}{8} \approx 0.375 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/8)}{\sin(\pi/4)} \approx 1.31$



{5, 7}

Regular pentagons · 7 meeting at every vertex

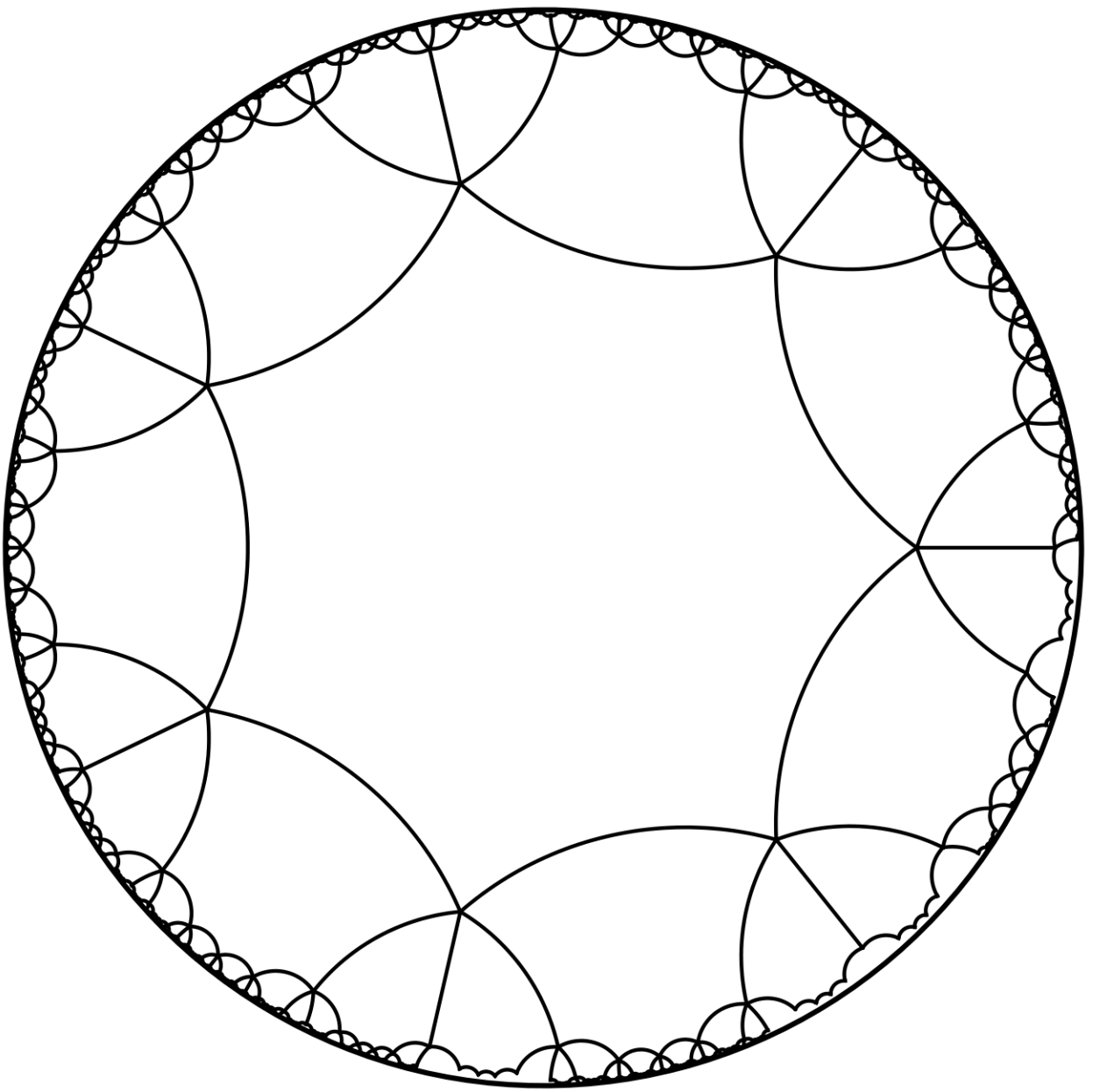
$\frac{1}{5} + \frac{1}{7} \approx 0.343 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/7)}{\sin(\pi/5)} \approx 1.53$



{6, 6}

Regular hexagons · 6 meeting at every vertex

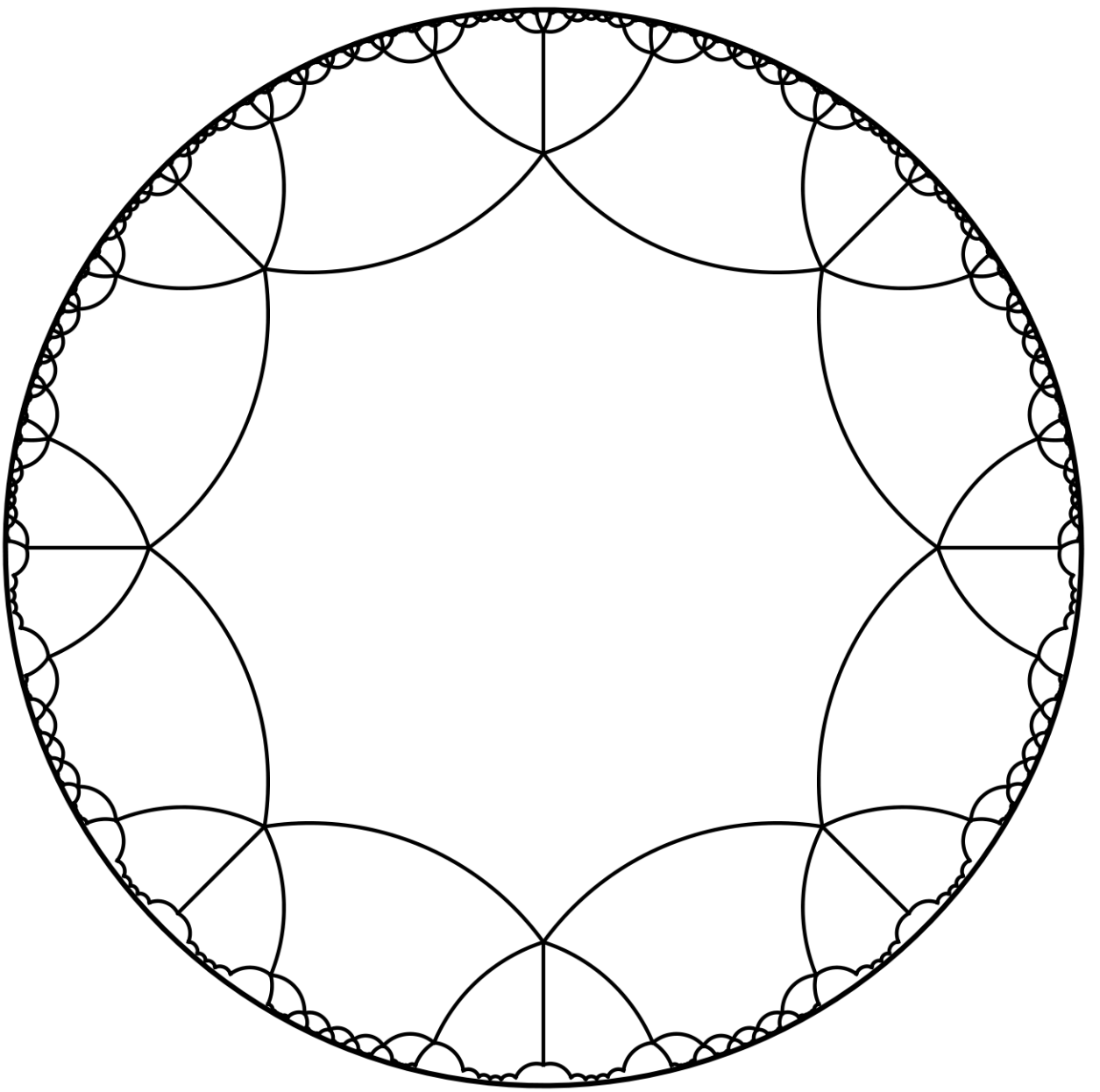
$\frac{1}{6} + \frac{1}{6} \approx 0.333 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/6)}{\sin(\pi/6)} \approx 1.73$



{7, 5}

Regular heptagons · 5 meeting at every vertex

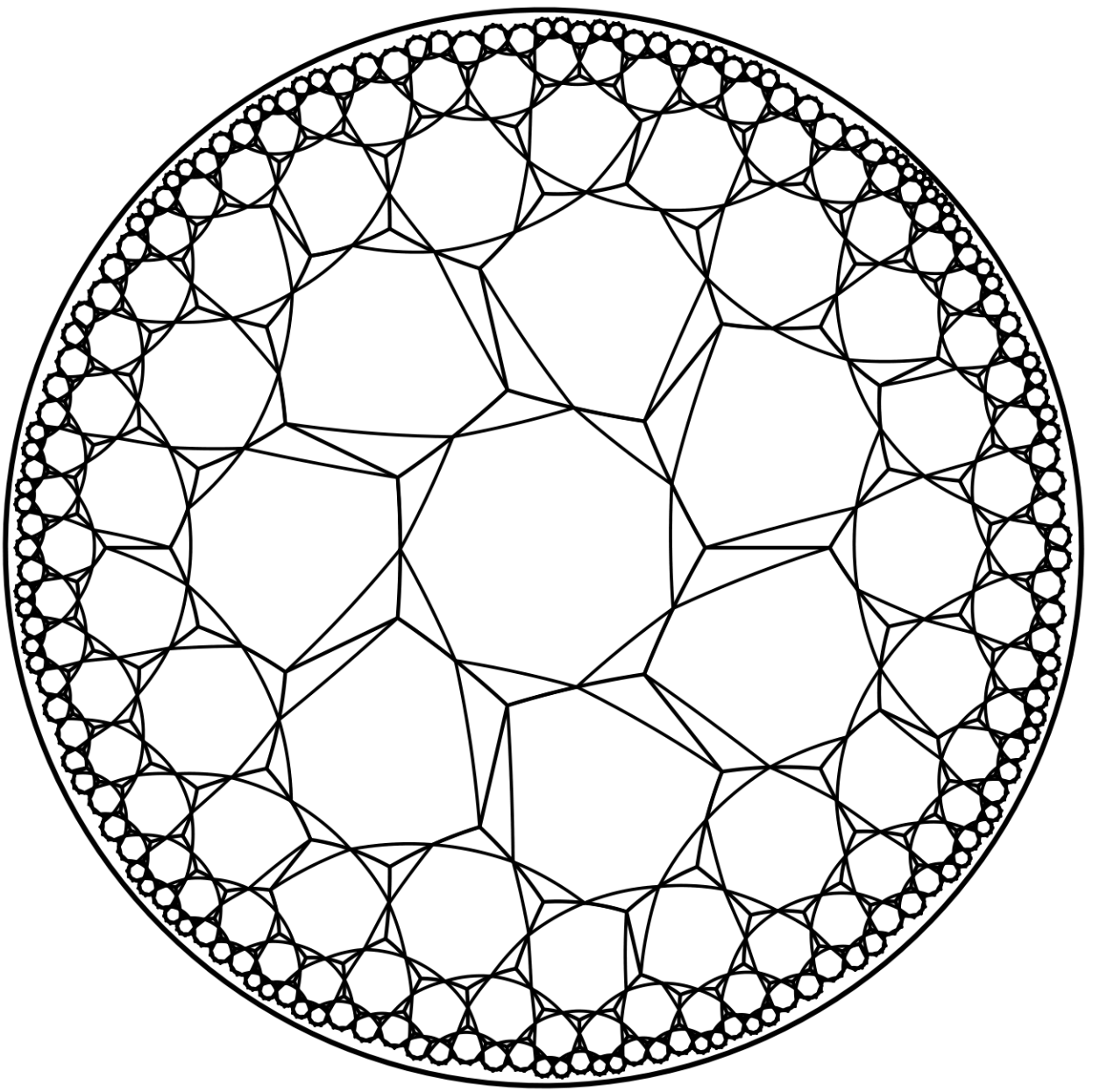
$\frac{1}{7} + \frac{1}{5} \approx 0.343 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/5)}{\sin(\pi/7)} \approx 1.86$



{8, 5}

Regular octagons · 5 meeting at every vertex

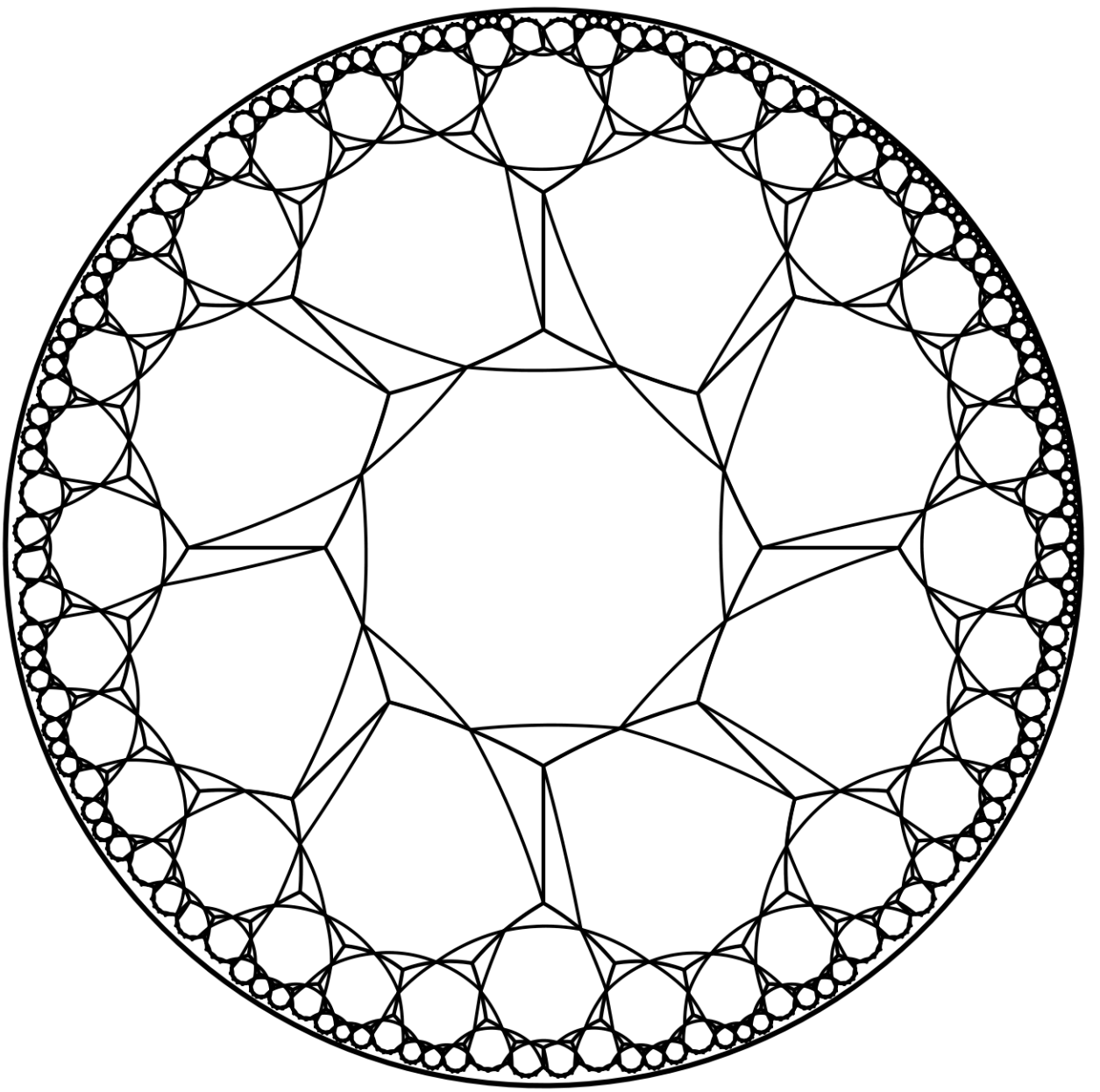
$\frac{1}{8} + \frac{1}{5} \approx 0.325 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/5)}{\sin(\pi/8)} \approx 2.11$



{7, 3} · nested

Regular heptagons · 3 meeting at every vertex

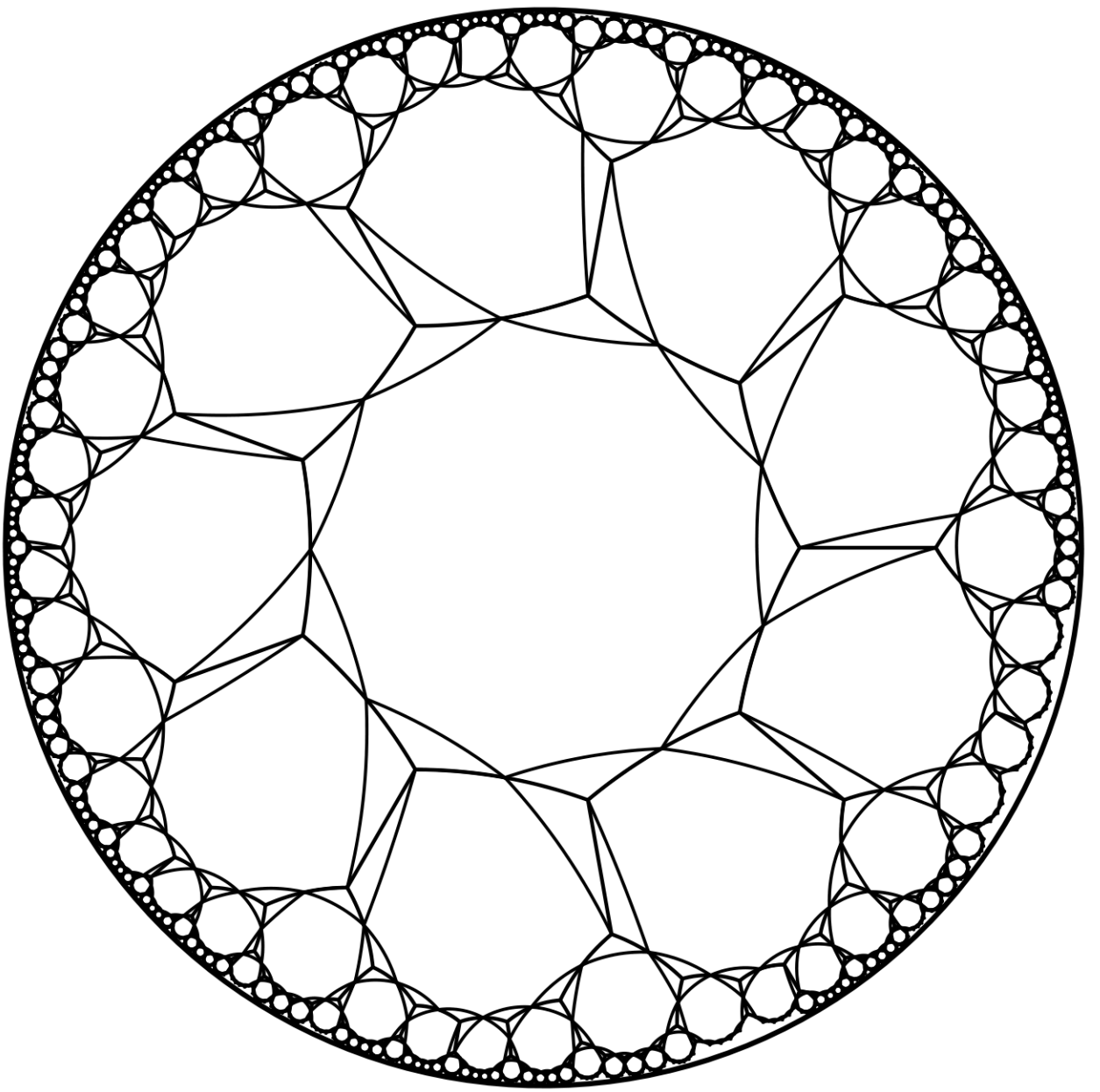
$\frac{1}{7}+\frac{1}{3}\approx 0.476<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/3)}{\sin(\pi/7)}\approx 1.15$



{8, 3} · nested

Regular octagons · 3 meeting at every vertex

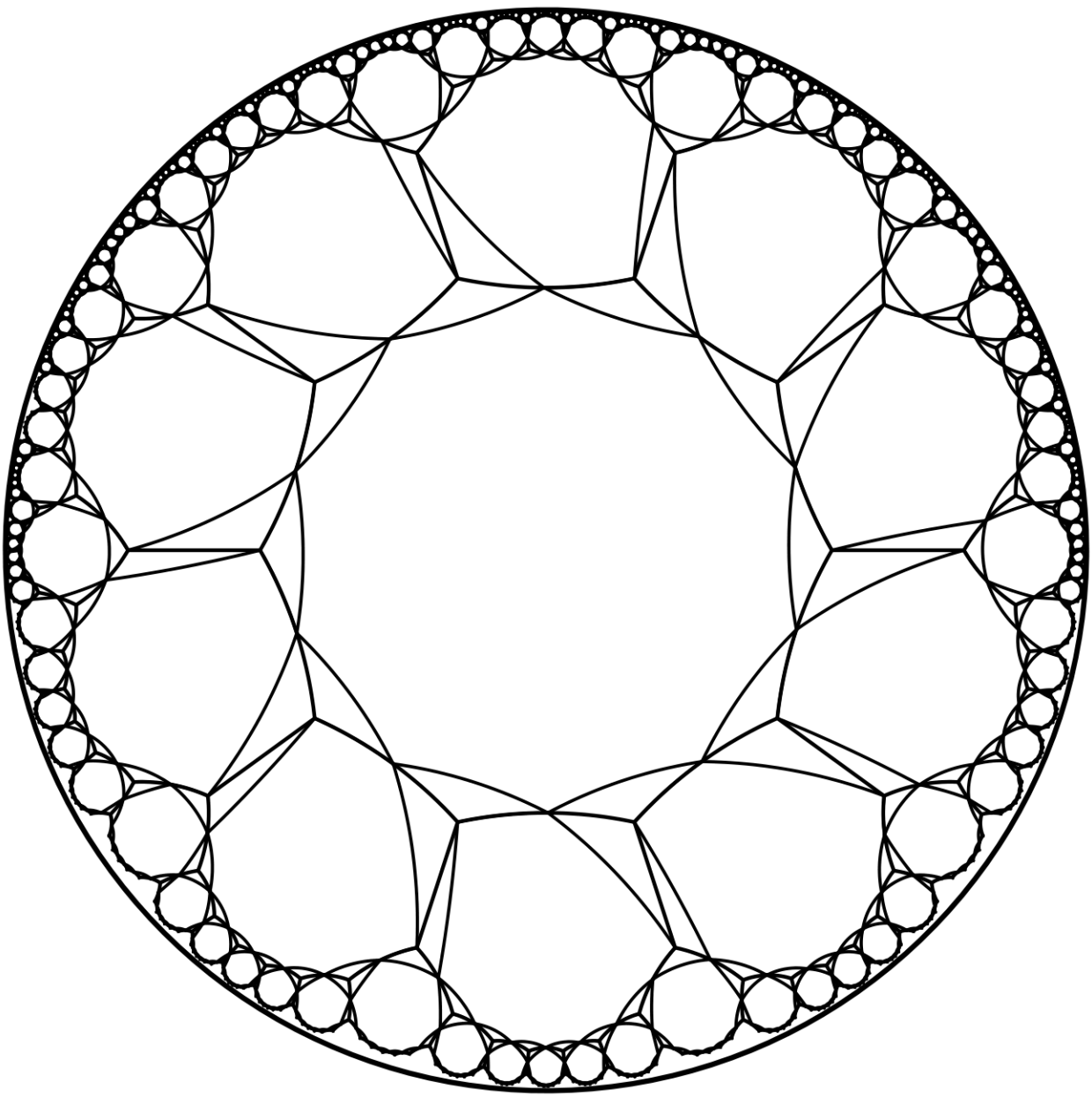
$\frac{1}{8}+\frac{1}{3}\approx 0.458<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/3)}{\sin(\pi/8)}\approx 1.31$



{9, 3} · nested

Regular nonagons · 3 meeting at every vertex

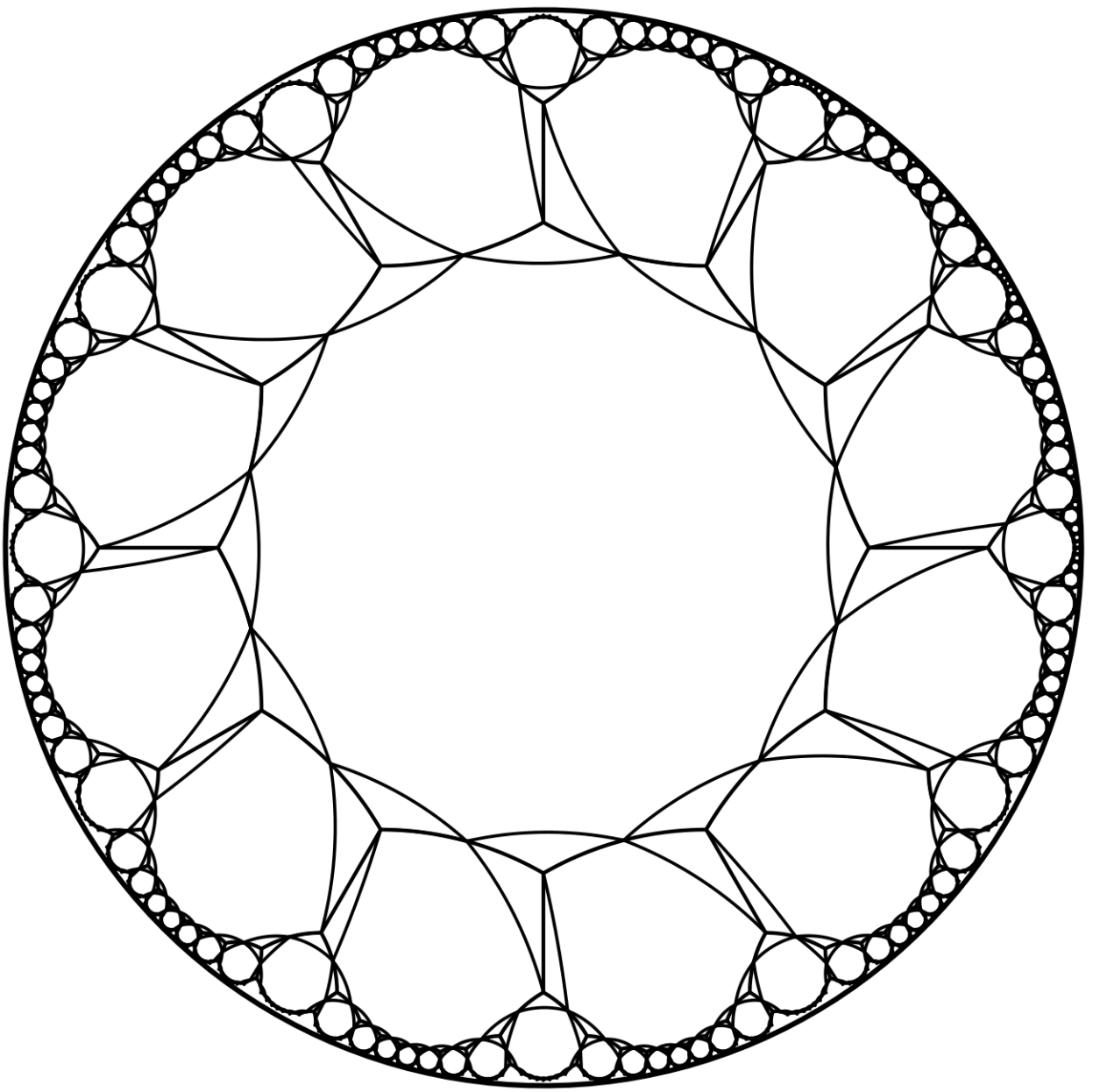
$\frac{1}{9}+\frac{1}{3}\approx 0.444<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/3)}{\sin(\pi/9)}\approx 1.46$



{10, 3} · nested

Regular decagons · 3 meeting at every vertex

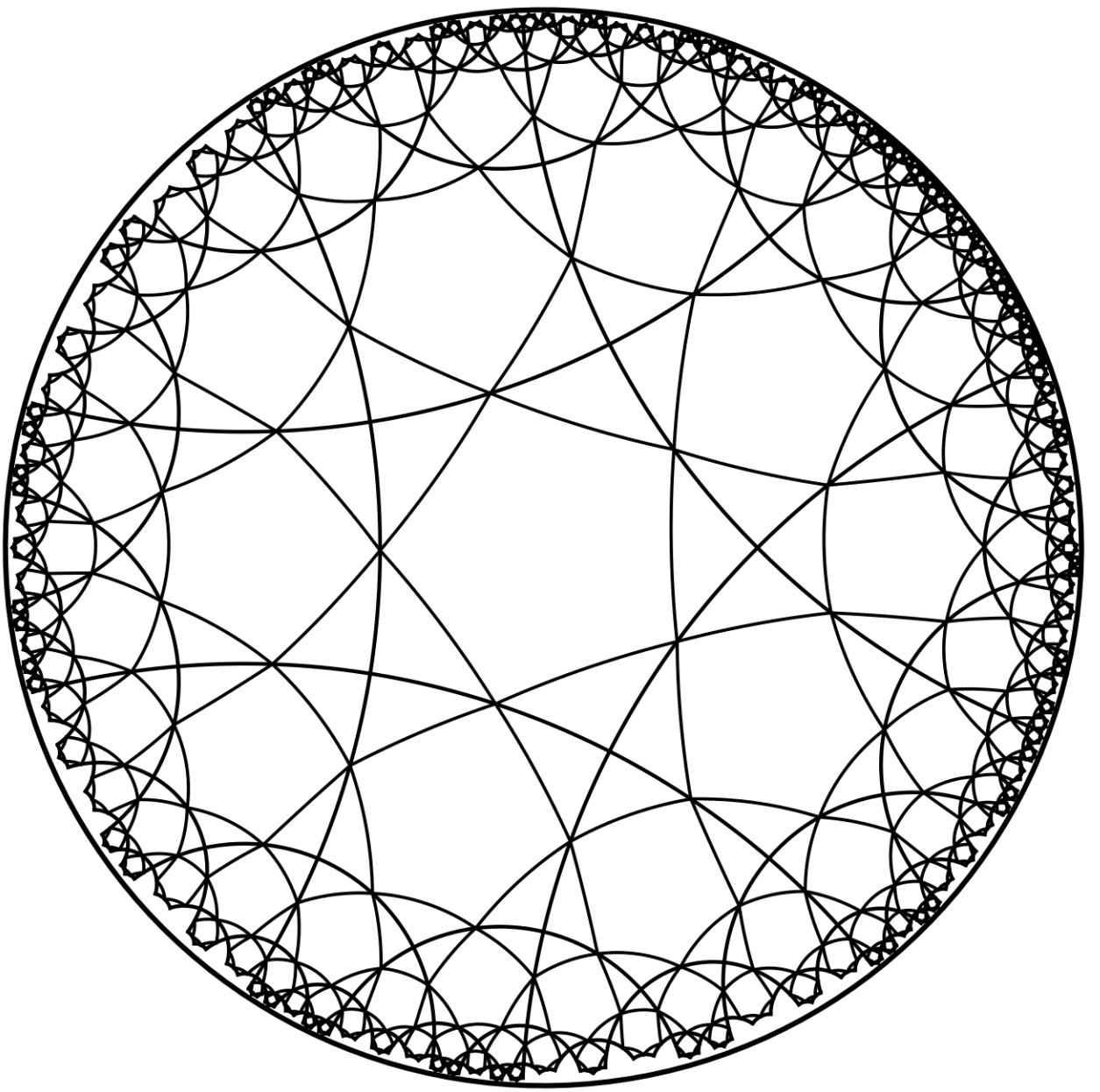
$\frac{1}{10} + \frac{1}{3} \approx 0.433 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/3)}{\sin(\pi/10)} \approx 1.62$



{12, 3} · nested

Regular dodecagons · 3 meeting at every vertex

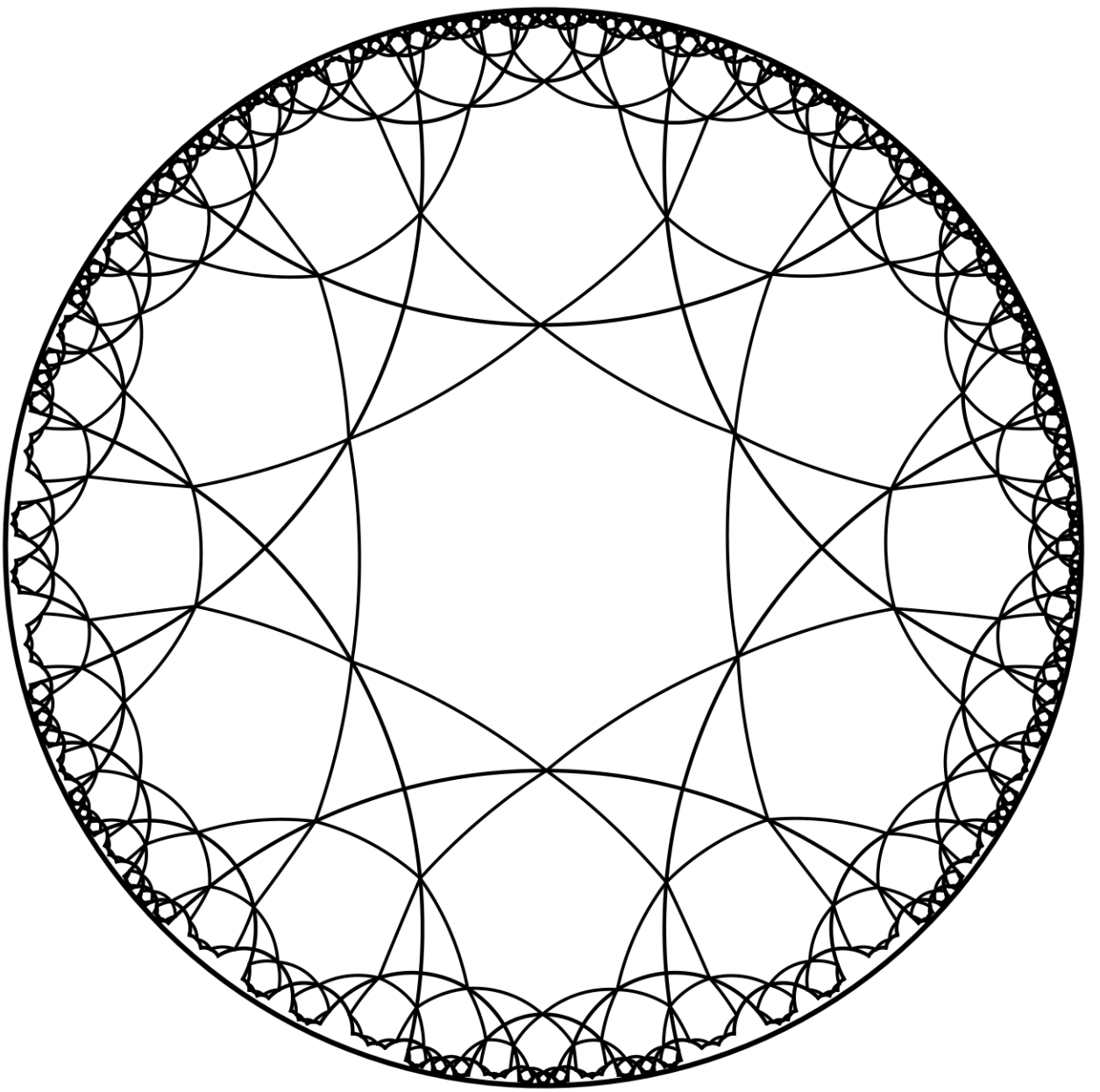
$\frac{1}{12} + \frac{1}{3} \approx 0.417 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/3)}{\sin(\pi/12)} \approx 1.93$



{5, 4} · nested

Regular pentagons · 4 meeting at every vertex

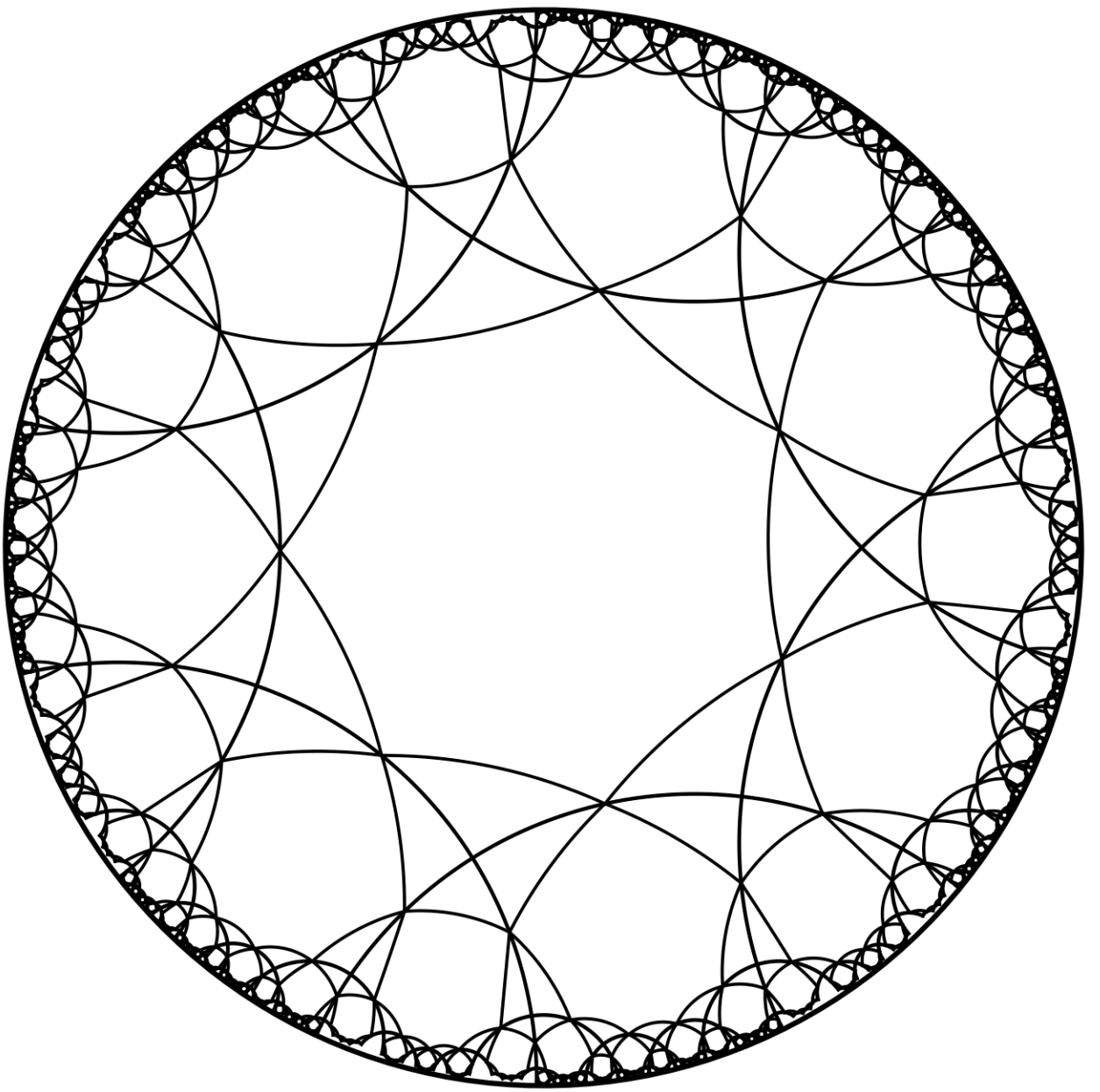
$\frac{1}{5}+\frac{1}{4}\approx 0.450<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/4)}{\sin(\pi/5)}\approx 1.20$



{6, 4} · nested

Regular hexagons · 4 meeting at every vertex

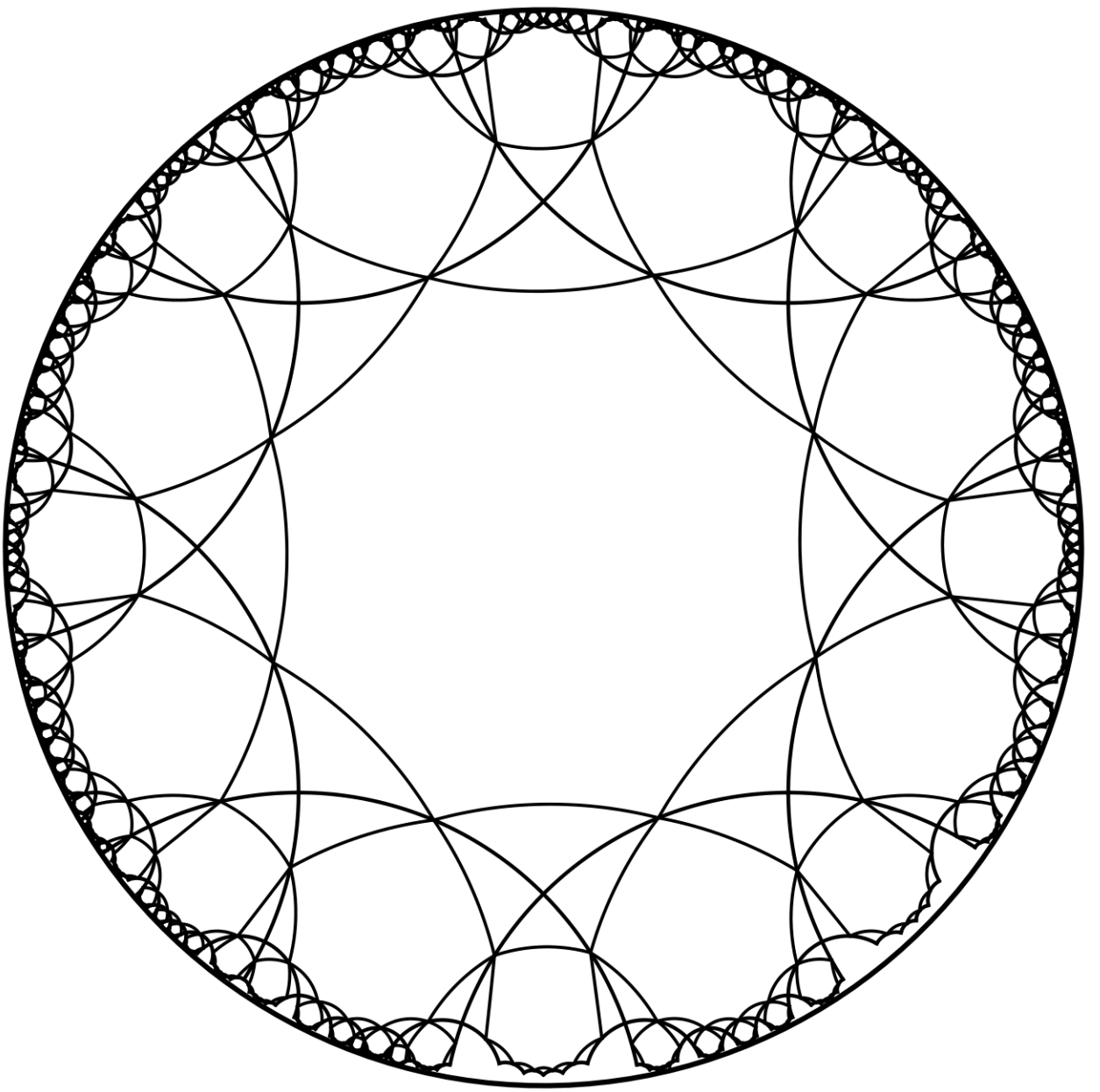
$\frac{1}{6}+\frac{1}{4}\approx 0.417<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/4)}{\sin(\pi/6)}\approx 1.41$



{7, 4} · nested

Regular heptagons · 4 meeting at every vertex

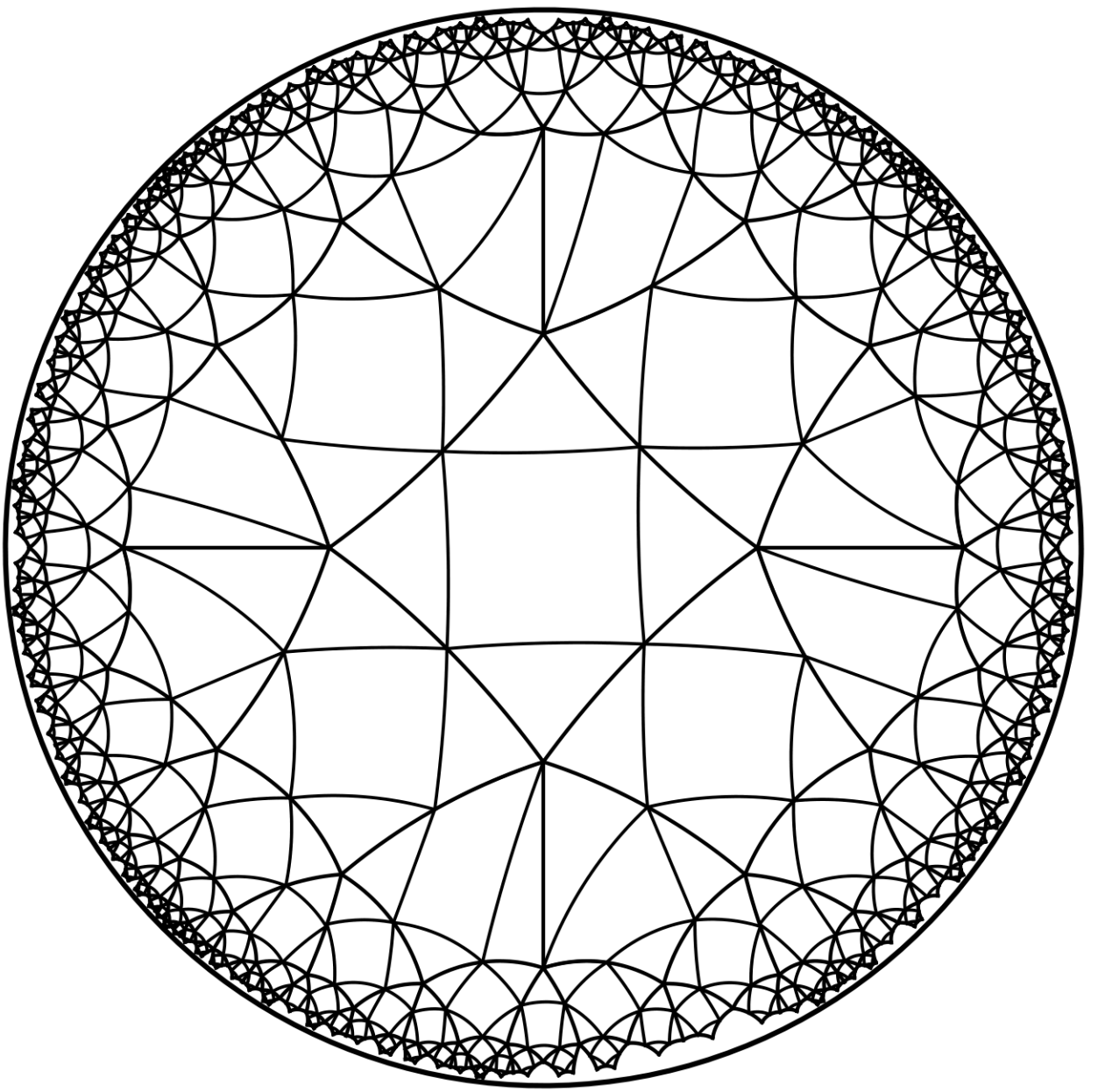
$\frac{1}{7}+\frac{1}{4}\approx 0.393<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/4)}{\sin(\pi/7)}\approx 1.63$



{8, 4} · nested

Regular octagons · 4 meeting at every vertex

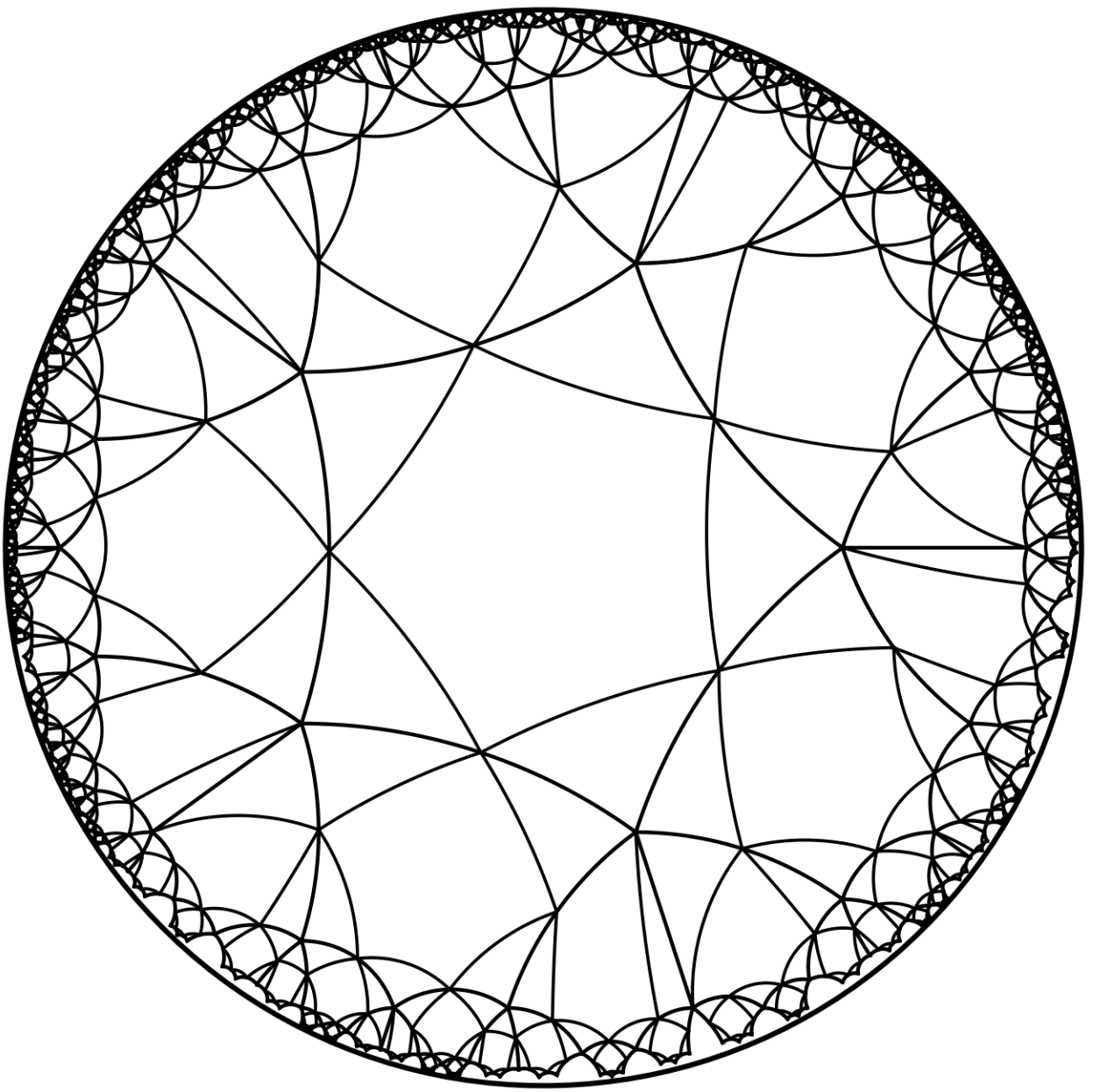
$\frac{1}{8} + \frac{1}{4} \approx 0.375 < \frac{1}{2}$ $\qquad \cosh R = \frac{\cos(\pi/4)}{\sin(\pi/8)} \approx 1.85$



{4, 5} · nested

Regular squares · 5 meeting at every vertex

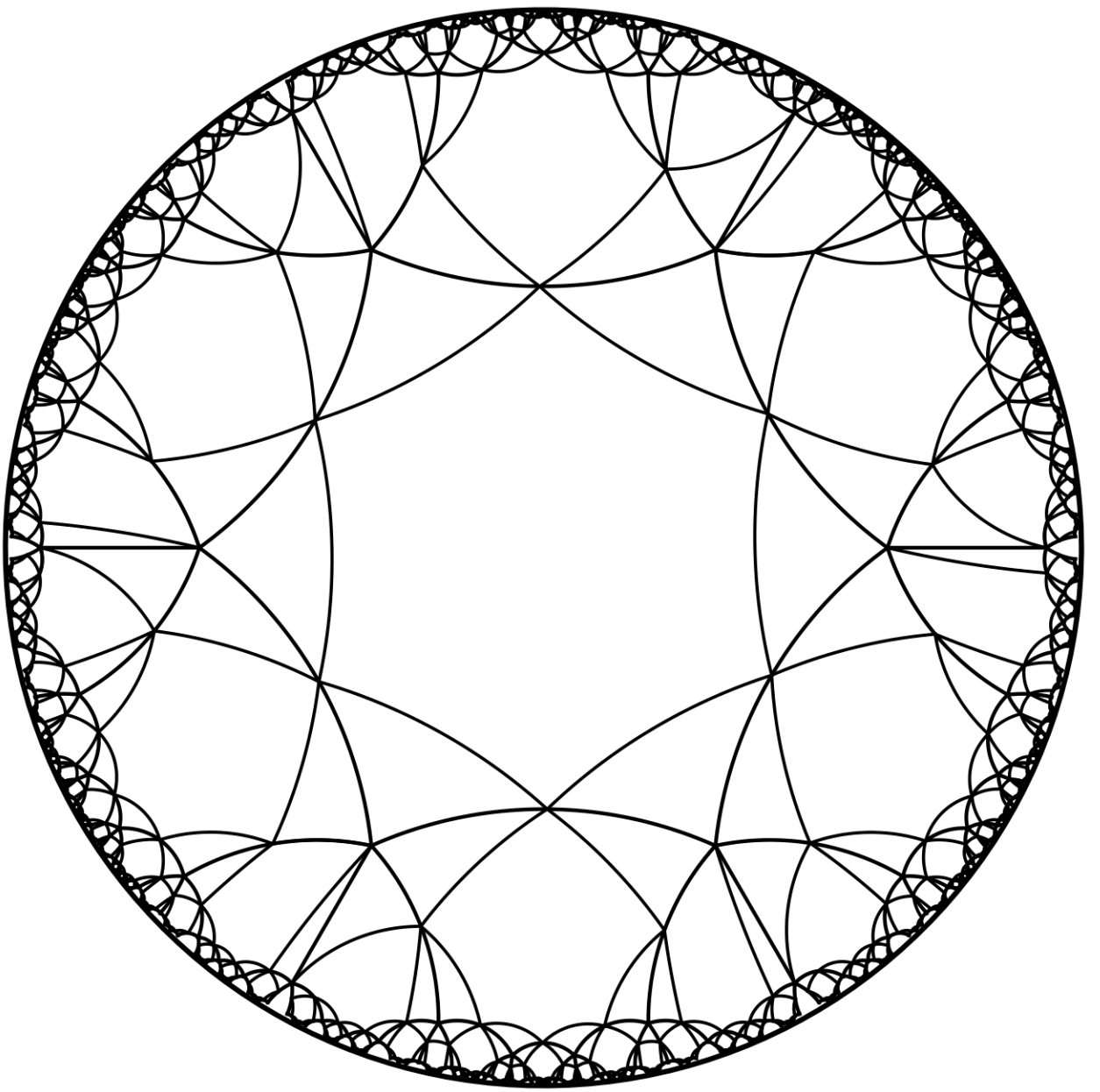
$\frac{1}{4}+\frac{1}{5}\approx 0.450<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/5)}{\sin(\pi/4)}\approx 1.14$



{5, 5} · nested

Regular pentagons · 5 meeting at every vertex

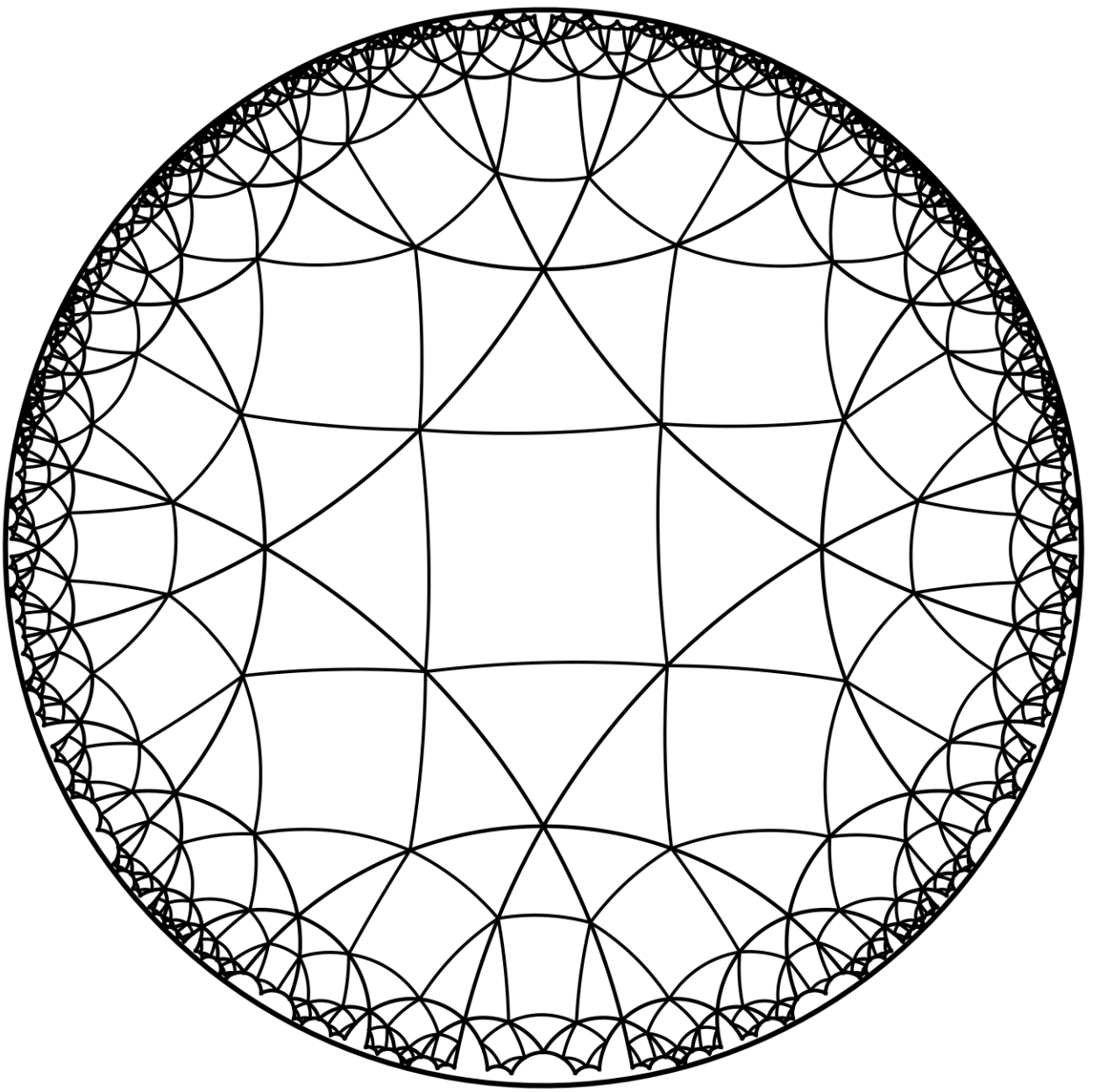
$\frac{1}{5}+\frac{1}{5}\approx 0.400<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/5)}{\sin(\pi/5)}\approx 1.38$



{6, 5} · nested

Regular hexagons · 5 meeting at every vertex

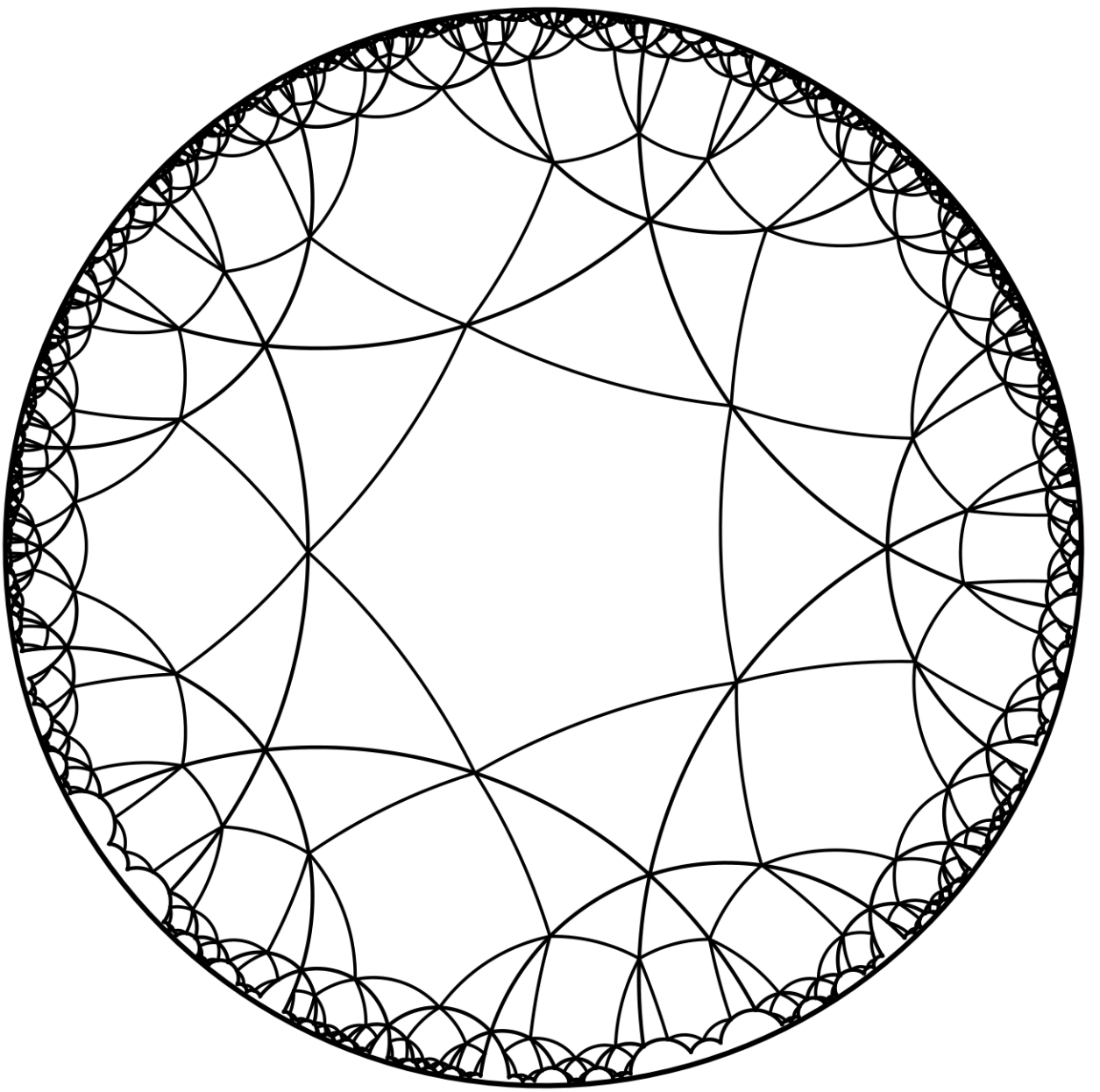
$\frac{1}{6}+\frac{1}{5}\approx 0.367<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/5)}{\sin(\pi/6)}\approx 1.62$



{4, 6} · nested

Regular squares · 6 meeting at every vertex

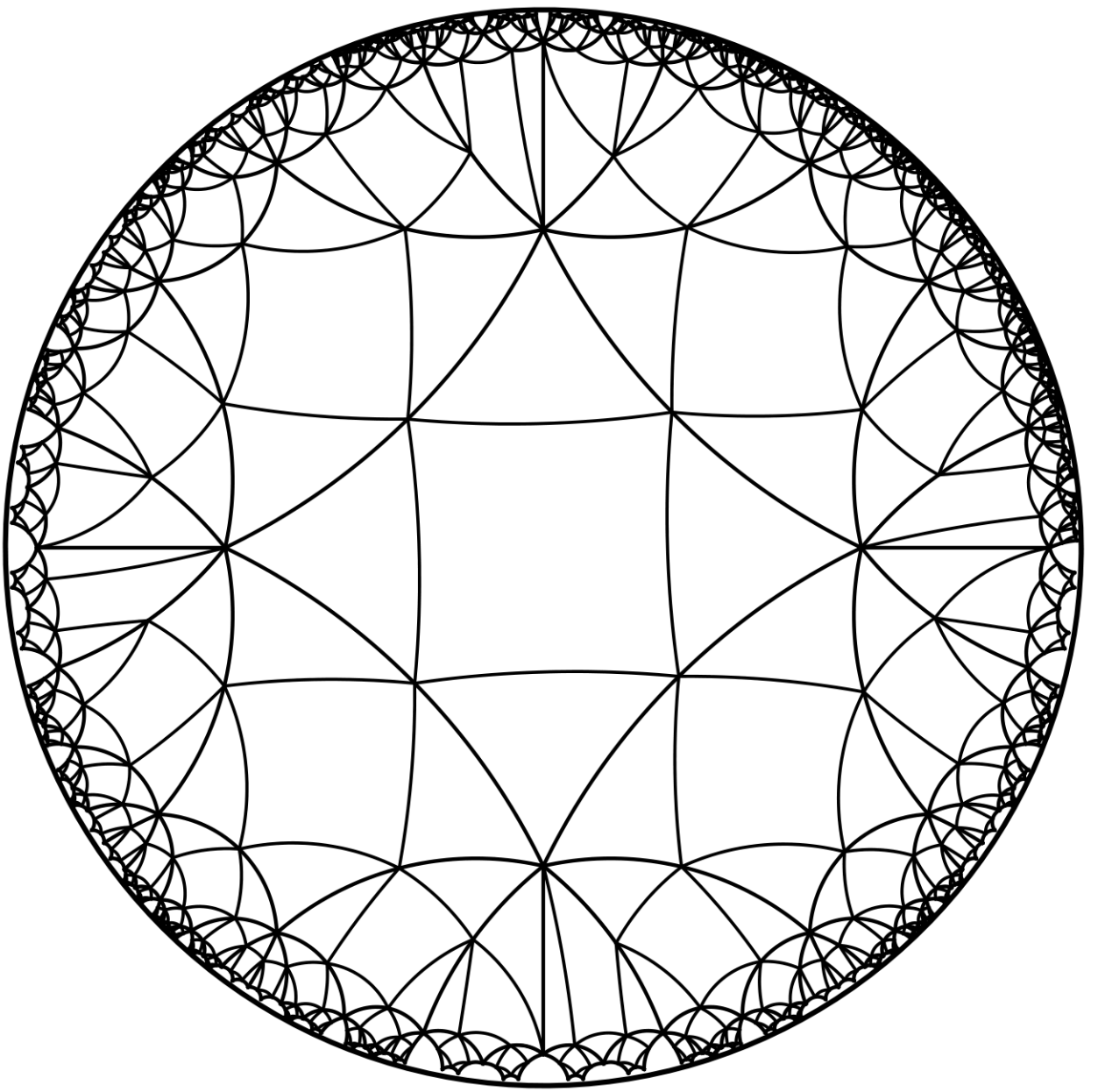
$\frac{1}{4}+\frac{1}{6}\approx 0.417<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/6)}{\sin(\pi/4)}\approx 1.22$



{5, 6} · nested

Regular pentagons · 6 meeting at every vertex

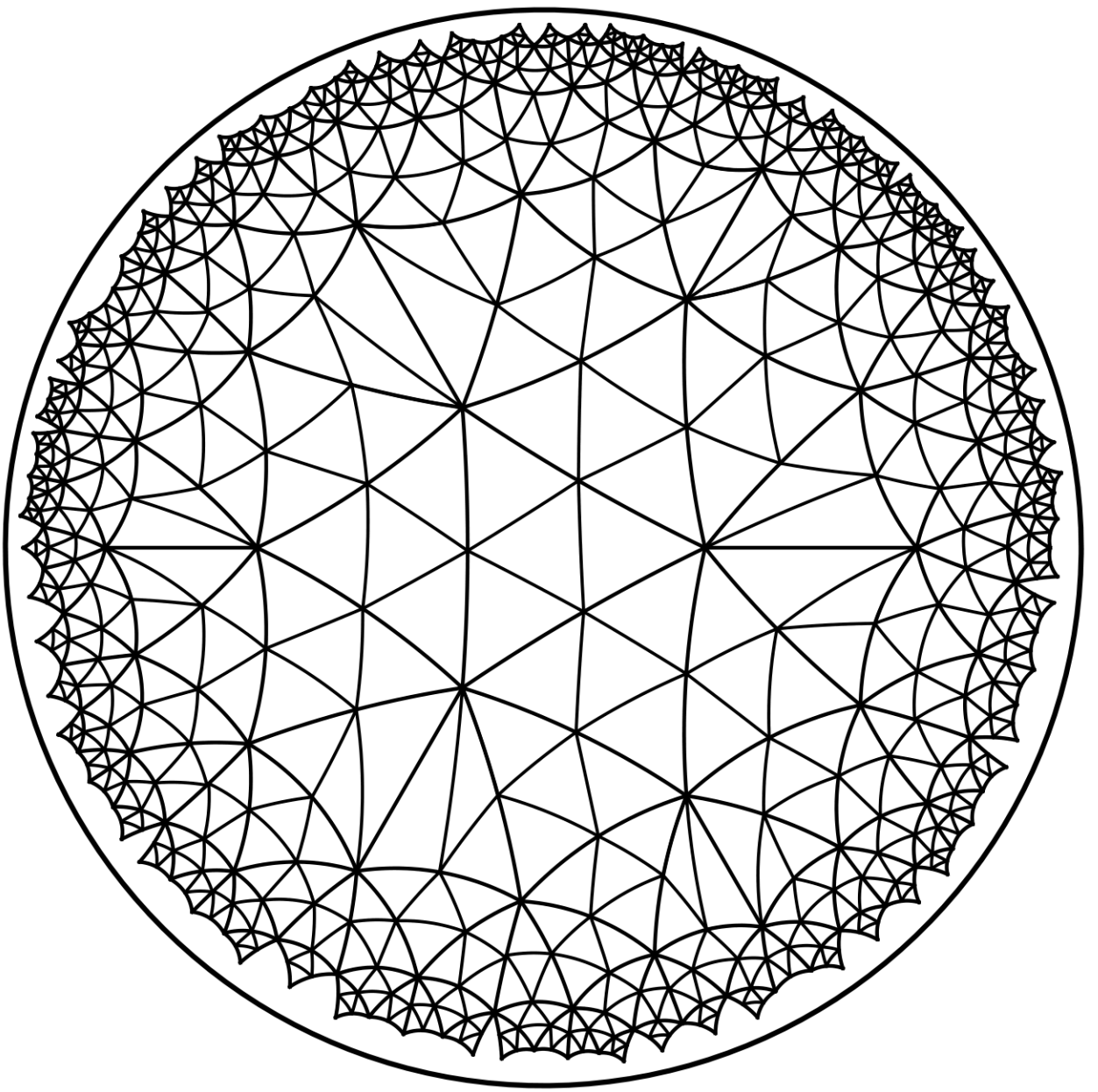
$\frac{1}{5}+\frac{1}{6}\approx 0.367<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/6)}{\sin(\pi/5)}\approx 1.47$



{4, 7} · nested

Regular squares · 7 meeting at every vertex

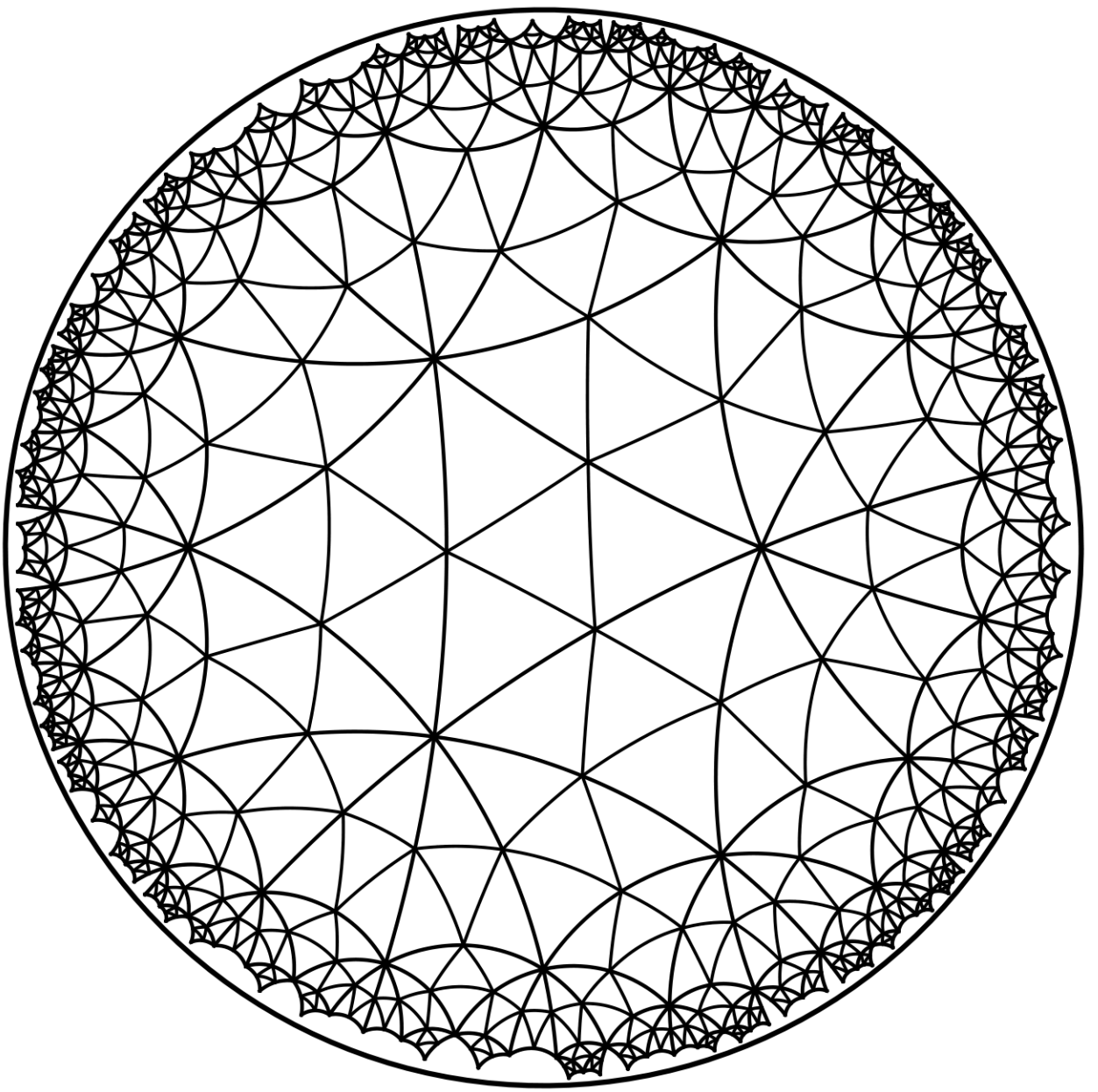
$\frac{1}{4} + \frac{1}{7} \approx 0.393 < \frac{1}{2}$ $\qquad \cosh R = \frac{\cos(\pi/7)}{\sin(\pi/4)} \approx 1.27$



{3, 7} · nested

Regular triangles · 7 meeting at every vertex

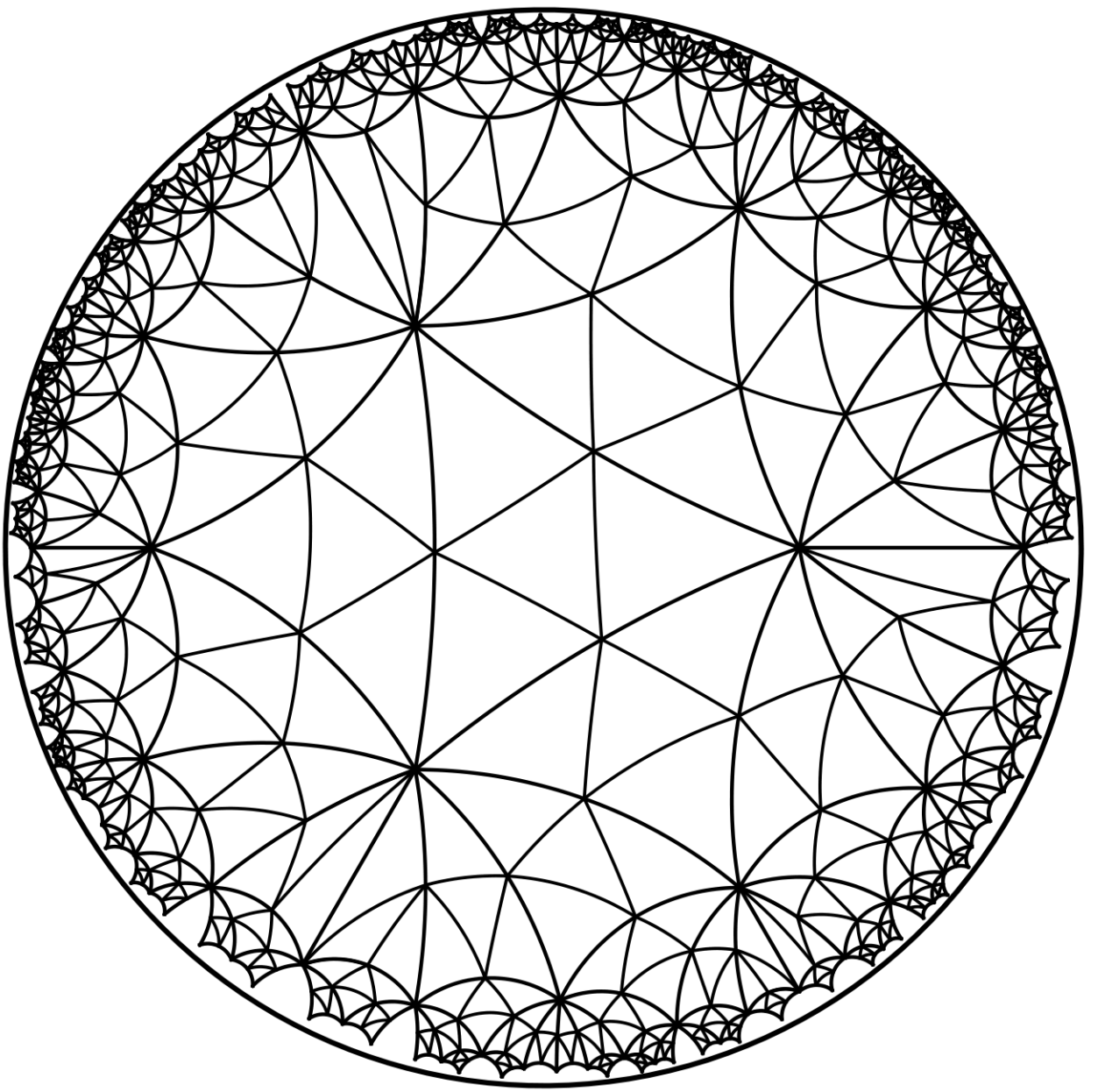
$\frac{1}{3}+\frac{1}{7}\approx 0.476<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/7)}{\sin(\pi/3)}\approx 1.04$



{3, 8} · nested

Regular triangles · 8 meeting at every vertex

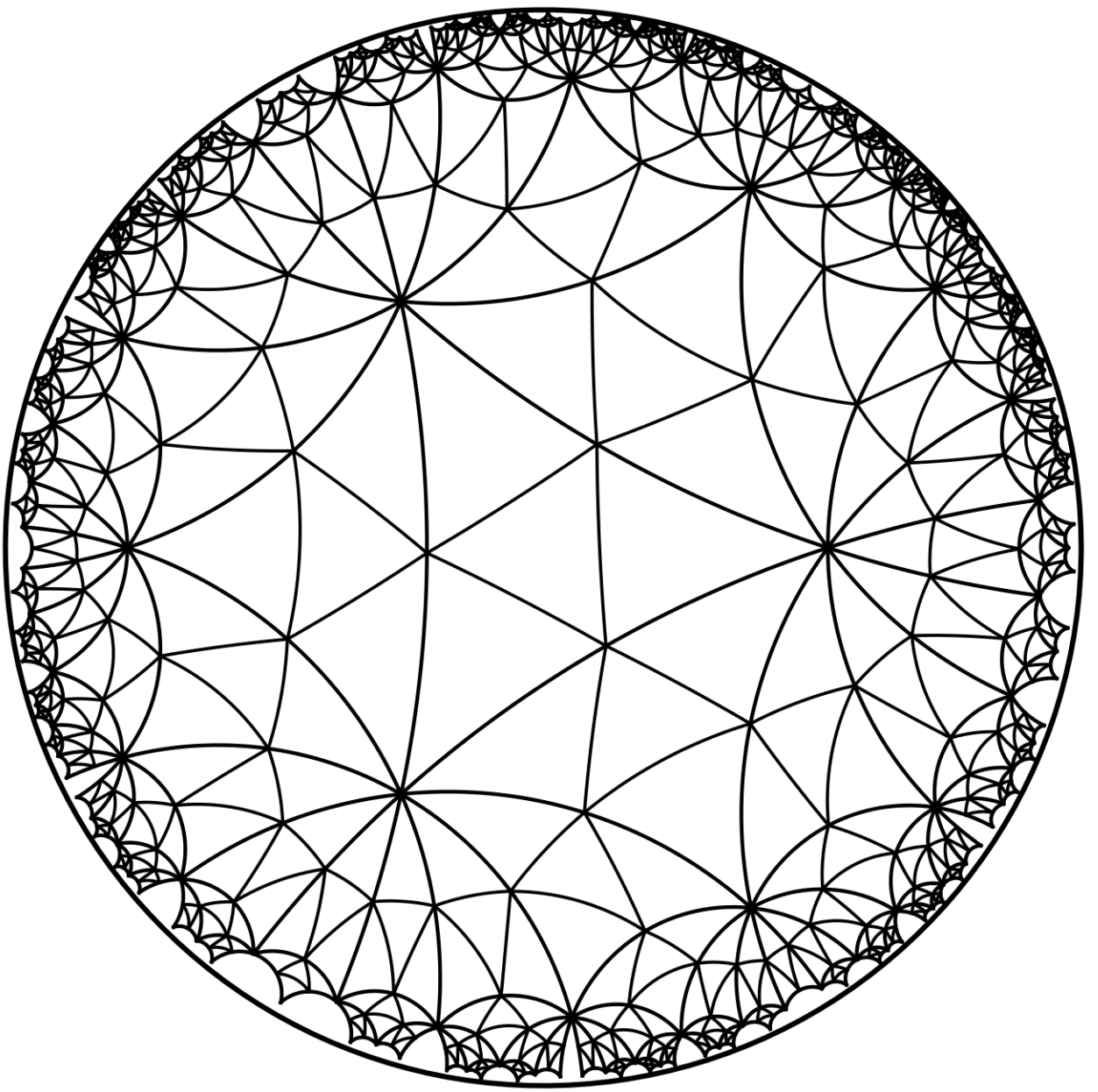
$\frac{1}{3}+\frac{1}{8}\approx 0.458<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/8)}{\sin(\pi/3)}\approx 1.07$



{3, 9} · nested

Regular triangles · 9 meeting at every vertex

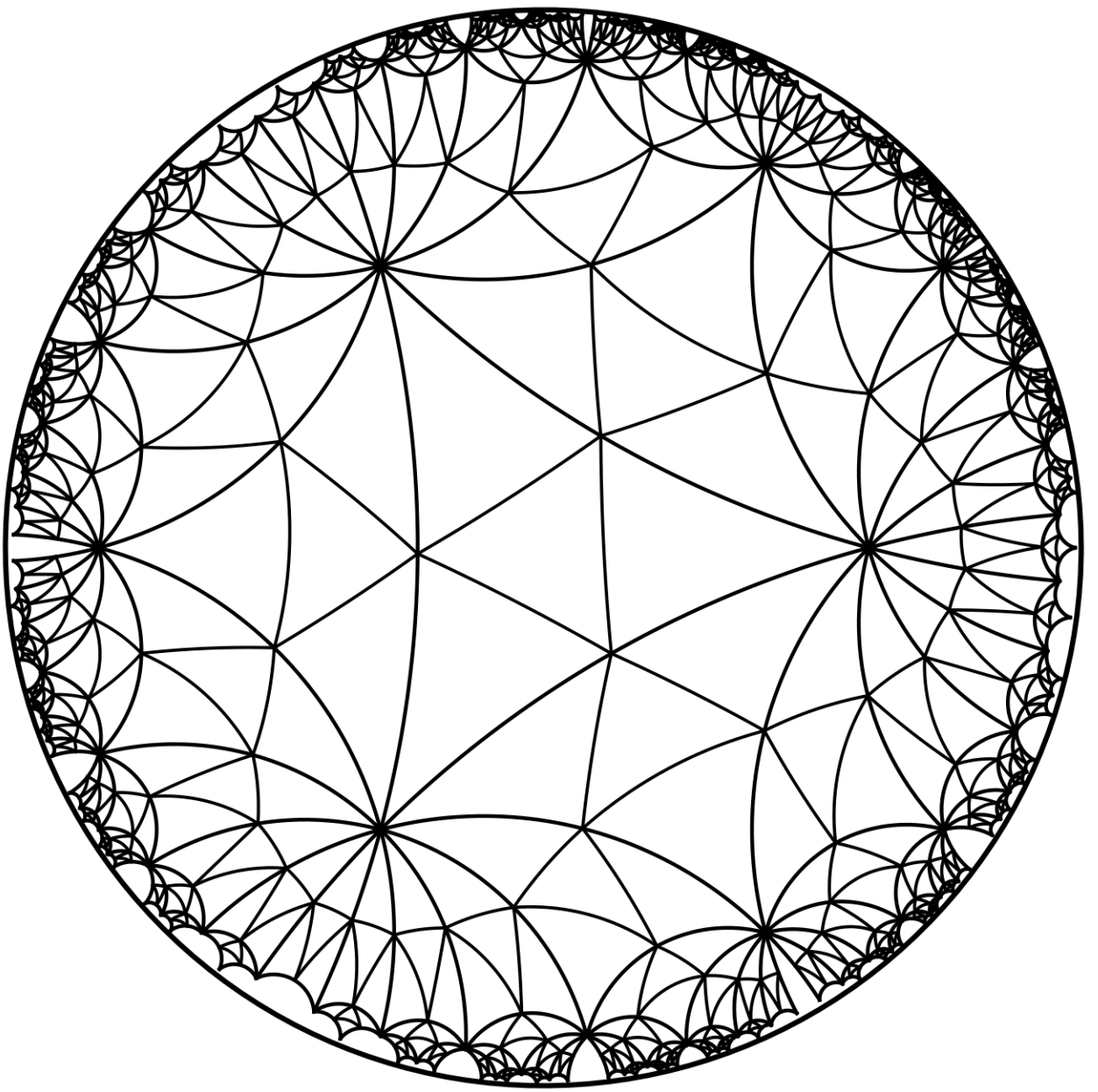
$\frac{1}{3} + \frac{1}{9} \approx 0.444 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/9)}{\sin(\pi/3)} \approx 1.09$



{3, 10} · nested

Regular triangles · 10 meeting at every vertex

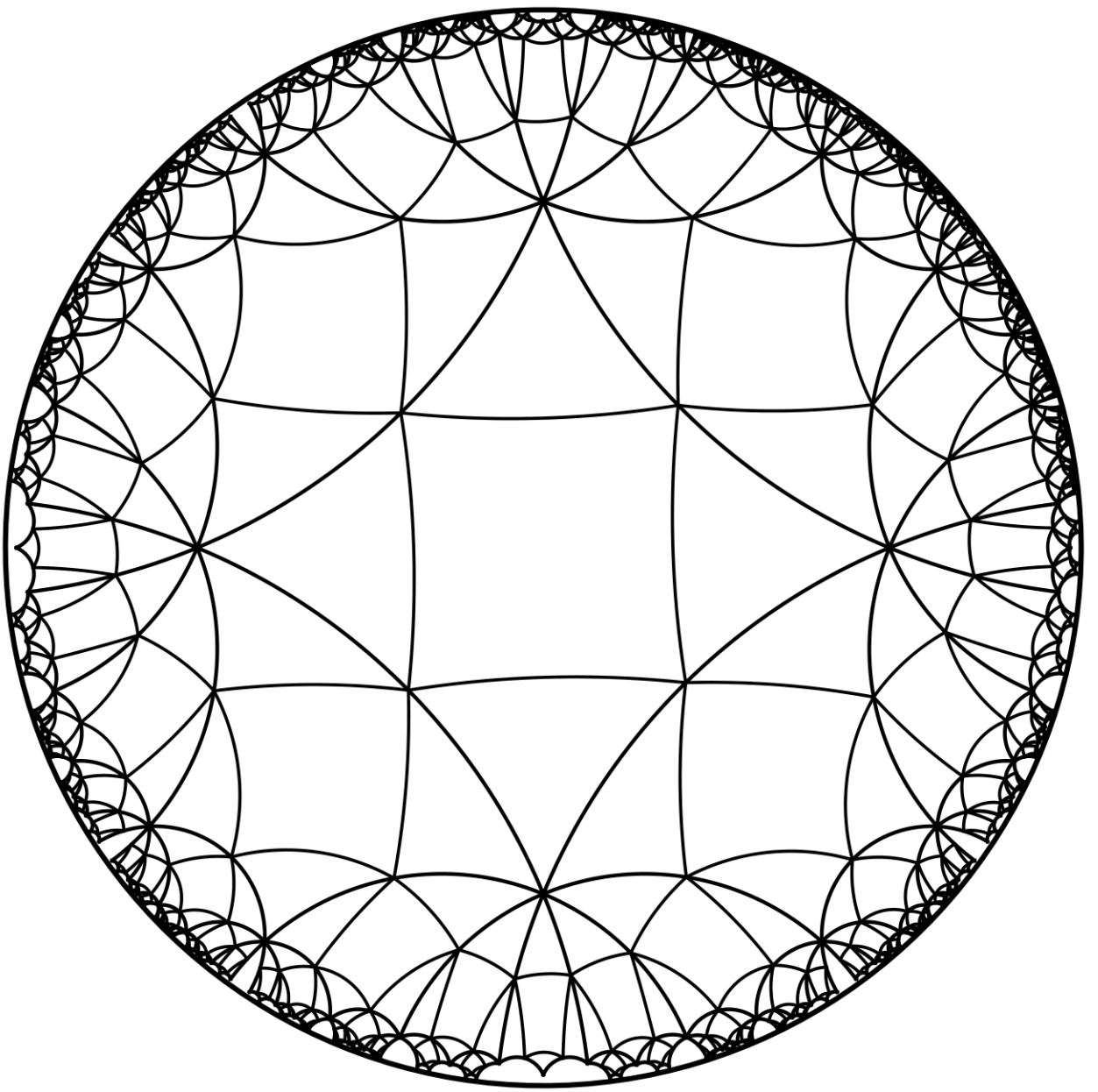
$\frac{1}{3} + \frac{1}{10} \approx 0.433 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/10)}{\sin(\pi/3)} \approx 1.10$



{3, 12} · nested

Regular triangles · 12 meeting at every vertex

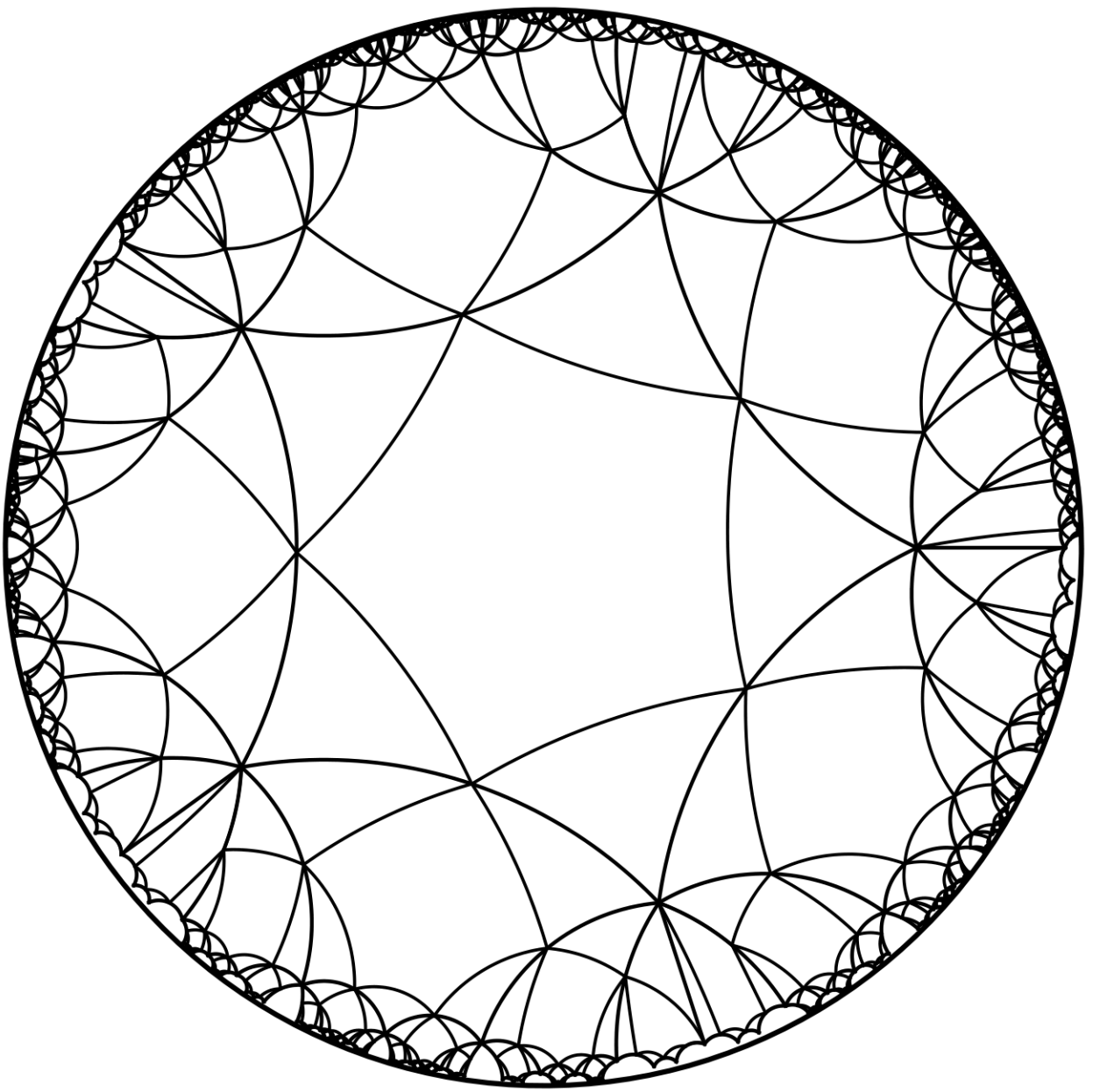
$\frac{1}{3} + \frac{1}{12} \approx 0.417 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/12)}{\sin(\pi/3)} \approx 1.12$



{4, 8} · nested

Regular squares · 8 meeting at every vertex

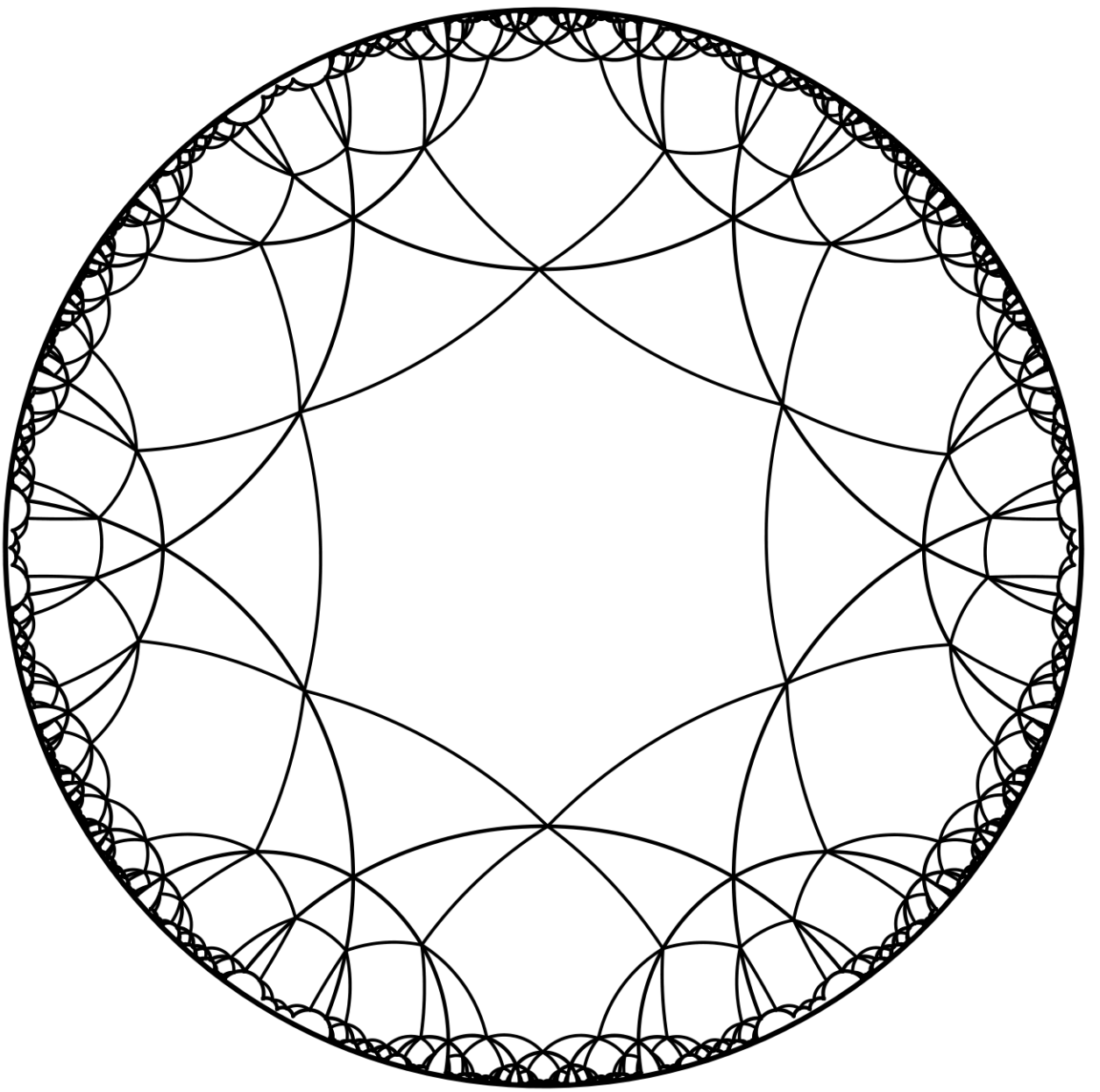
$\frac{1}{4}+\frac{1}{8}\approx 0.375<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/8)}{\sin(\pi/4)}\approx 1.31$



{5, 7} · nested

Regular pentagons · 7 meeting at every vertex

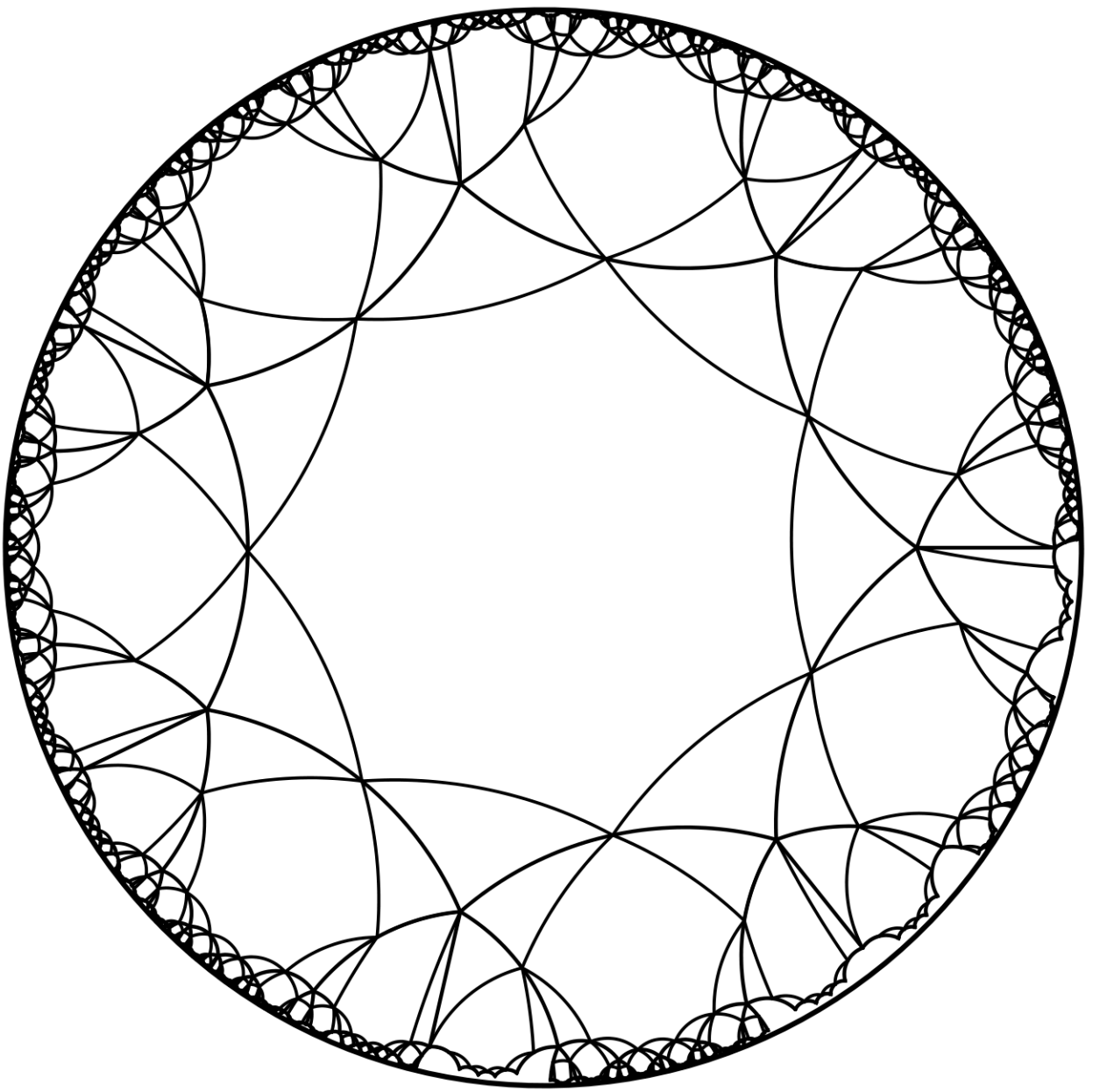
$\frac{1}{5}+\frac{1}{7}\approx 0.343<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/7)}{\sin(\pi/5)}\approx 1.53$



{6, 6} · nested

Regular hexagons · 6 meeting at every vertex

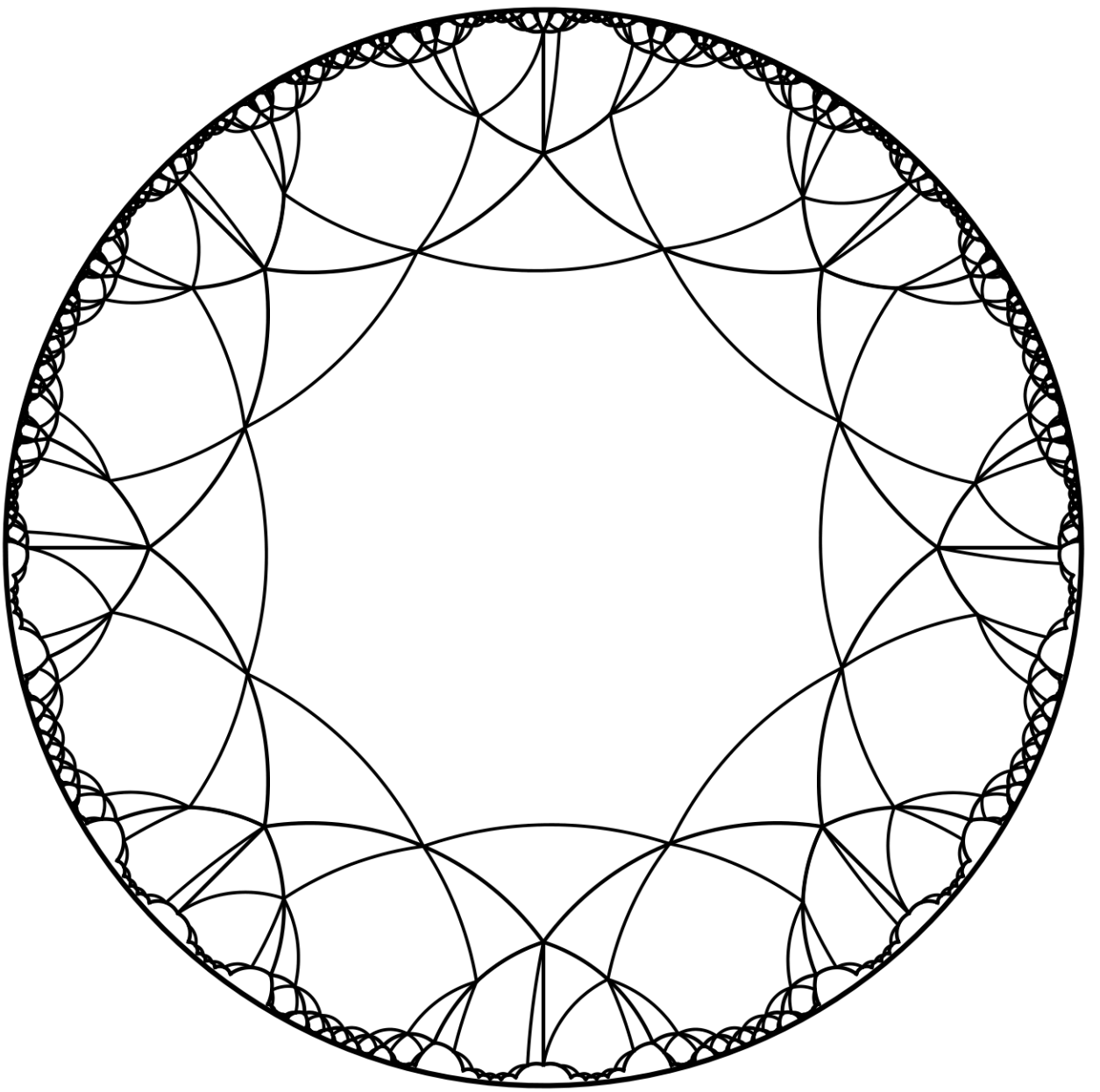
$\frac{1}{6}+\frac{1}{6}\approx 0.333<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/6)}{\sin(\pi/6)}\approx 1.73$



{7, 5} · nested

Regular heptagons · 5 meeting at every vertex

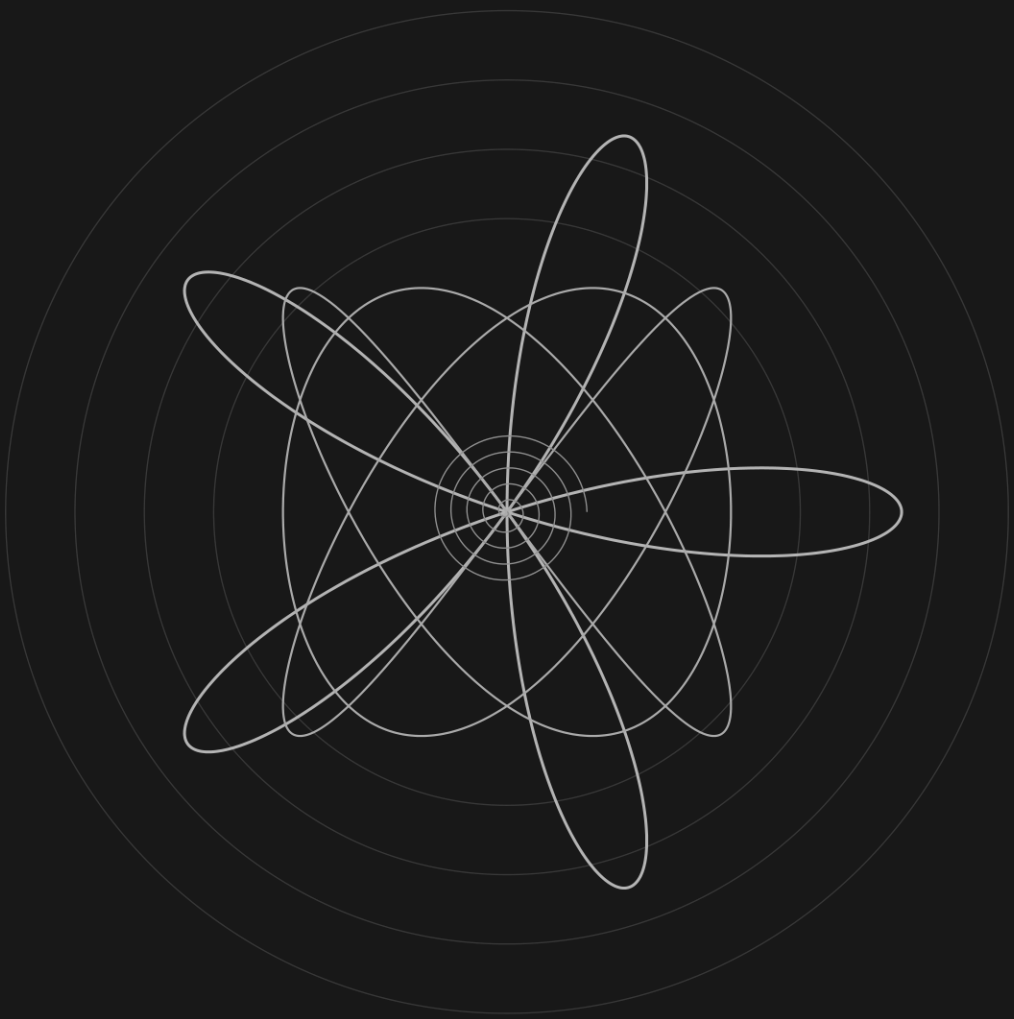
$\frac{1}{7}+\frac{1}{5}\approx 0.343<\frac{1}{2}$ $\qquad$ $\cosh R=\frac{\cos(\pi/5)}{\sin(\pi/7)}\approx 1.86$



{8, 5} · nested

Regular octagons · 5 meeting at every vertex

$\frac{1}{8} + \frac{1}{5} \approx 0.325 < \frac{1}{2}$ $\qquad$ $\cosh R = \frac{\cos(\pi/5)}{\sin(\pi/8)} \approx 2.11$



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